EVIDENCE OF DUST GRAIN GROWTH IN YOUNG PROTOSTELLAR DISK IRS 63 FOUND USING POLARIZED LIGHT

by

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A thesis submitted to the
Graduate Program in the Department of Physics, Engineering Physics & Astronomy
in conformity with the requirements for
the degree of Master of Science

Queen's University
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Dedication

Dedication goes here...

[**ADD:** Here is how I will add the text on things I need to add]

Q: Here is how I add text where I have a question

[NOTE: Here is how I add text on a note I have]

"Text in this format is when I directly copy text from a source, and it should be followed by the source it was from. This text is yet to be put into my own words. "

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*Make sure when (ref1, ref2) its in chronological order?

SC: Simon Coude, Pierre Bastien,: If there is a way to abridge the author list, I suggest doing that! *[NOTE: I still need to address this comment]*

Abstract

To understand how planets form, we must study how dust grains in protostellar disks grow from micron-sized grains into larger objects. An important part of understanding this process is determining the sizes of the dust grains in young disks. Constraining dust grain sizes is challenging, as grains are small, far away, and cannot be measured directly. Polarized light from dust self-scattering provides an opportunity to constrain dust grain sizes, as the resulting polarized light depends on dust grain size, disk geometry, and optical depth.

This thesis analyzes multi-wavelength Atacama Large Millimeter/submillimeter Array (ALMA) polarization emission observations of the Class I protostellar disk IRS 63 at 2.1 mm, 1.5 mm, 1.3 mm, and 0.87 mm (ALMA Bands 4, 5, 6, and 7). The polarization morphology transitions from an azimuthal pattern to a more uniform pattern aligned with the minor axis as wavelength decreases. This shift in pattern indicates a change in the dominant polarization mechanism, with dust self-scattering dominating at shorter wavelengths and in the inner part of the disk. Intensity profiles along the minor axis of the disk indicate that IRS 63 could be geometrically thick and flared.

Polarization observations of the central region of the disk are compared to models to constrain the maximum grain size. Both spherical and irregular grain shape models are considered, with irregular grains modelled using the neural network `glitterin` from **?**. Spherical grain modelling favours slightly porous grains with a maximum grain size of approximately 245 μm. Models of irregular grains favour a similar maximum grain size of approximately 212 μm. These grain sizes are much larger than typical interstellar medium dust grains ($\sim$ 0.1 μm) and cloud core grain sizes ($\sim$ 1 μm), providing evidence for dust grain growth in the young disk IRS 63. These results highlight how multi-wavelength polarization observations can be used to constrain dust grain size in young protostellar disks and provide evidence that dust grain growth is occurring during the early stage of planet formation.

Acknowledgements

Acknowledgments if any...

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Chapter 1

Introduction

[NOTE: The intro is in progress and does not need to be reviewed yet]

1.1 Star and Planet Formation

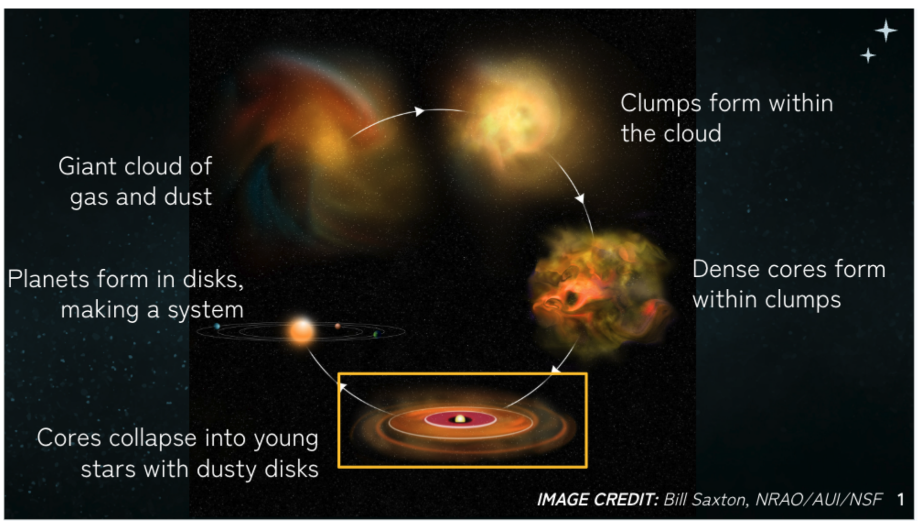

Figure 1.1: **[ADD:** caption] *[NOTE: If keeping, ask for permission]*

1.2 Protostellar Disks

The minor axis of the disk is 90° from the disk polarization angle (same as major axis direction, as shown in Figure ??).

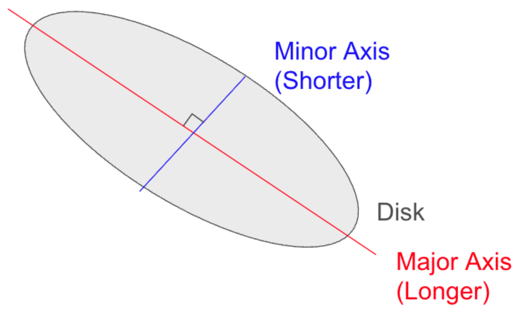

Figure 1.2: Schematic of the minor and major axes of a disk, highlighting their 90° offset. *[NOTE: If this figure is not necessary I will take it out. If it is useful I will make a better one!]*

1.3 Dust in Protostellar Disks

1.3.1 ISM Dust Grains

The disk's mass is 99% made up of gas, and the other 1% is dust (?). The dust that is now in the disk is originally from the interstellar medium (ISM), is the raw material that will make up terrestrial planets, and the cores of giant planets (?). Dust in the ISM is typically sub-micron in size (?) and made up of silicate, carbonaceous and icy particles (?).

1.3.2 Grain Growth

Although dust only accounts for 1% of the disks initial mass, its evolution plays a vital role in the evolution of the disk and the formation of planets (**?**). The first step in planet formation is called grain growth, sometimes called dust coagulation (**?**). Grain growth refers to the process where transport mechanisms in the disk lead to many grain-grain collisions, where the grains stick together and form a larger particle (**?**). During this phase, the grains begin as micron-sized particles and grow to centimetres in size and larger, and become planetesimals (**?**). This process is highlighted in the bottom right of Figure **??**.

Q: Should I go into more detail about how this happens? turbulent mixing, settling, drift, etc (**?**) *[NOTE: If so, there is a much better image in* **?** *that I can swap out for Figure* **??***]*.

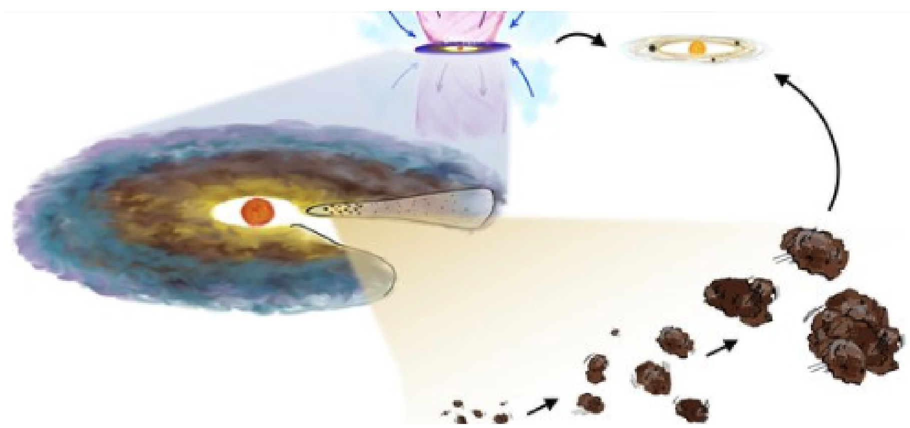

Figure 1.3: Schematic of dust grain growth, the first stage of planet formation. **[ADD:** reference] *[NOTE: If keeping this image, I will get a higher resolution version, and ask for permission.]*

1.3.3 Effects of Grain Size on Observed Emission

Dust grains dominate the opacity in the disk, controlling how radiation is absorbed, scattered and emitted (**?**). The opacity depends on several properties of the dust, including its shape, composition and size (**?**).

At millimetre wavelengths, the continuum emission from disks is dominated by thermal emission from the dust, which is controlled by the opacity of the dust (**?**). The efficiency with which the dust grains absorb and emit radiation depends on their size relative to the observing wavelength (**?**).

Grain size also affects the scattering properties of dust. Smaller grain sizes at millimetre wavelengths are less efficient at scattering light (**?**). However, larger grain sizes that are close to the observing wavelength are much more efficient at scattering light (**?**).

Because the absorption and scattering properties of dust grains depend on their size relative to the observing wavelength, observations at millimetre wavelengths can be used to study dust grain growth in disks (**?**).

1.4 Polarization in Protostellar Disks

1.4.1 Polarization

[NOTE: What is polarization]

[NOTE: Described by Stokes I, Q, U, V]

[NOTE: Only linear, ignore V]

[NOTE: Characterized by P(OLI?) and angle]

"Millimeter and submillimeter observations of continuum linear dust polarization provide insight into dust grain growth in protoplanetary disks, which are the progenitors of planetary systems" **?**

*"Polarized emission from dust on protoplanetary disks has been observed at millimeter and submillimeter wavelengths in an increasing number of sources"***?** *"Multiple mechanisms could theoretically produce this polarized emission, including the alignment of nonspherical dust grains to the magnetic field (e.g., Lazarian 2007) or radiation anisotropy (Tazaki et al. 2017), aerodynamic alignment due to gas–dust interactions (Gold 1952; Yang et al. 2019), or self-scattering of thermal dust emission (e.g., Kataoka et al. 2015; Yang et al. 2016, 2017). "***?**

*"Multiwavelength polarization observations are necessary to distinguish between these mechanisms and to extract information about dust properties from the polarized emission"***?**

*" the polarized emission seen in most disks does not match the pattern that would be produced by dust grains aligning to magnetic field morphologies predicted by theory. Instead, the polarization seen in many disks is better explained by other mechanisms."***?** *" In an inclined axisymmetric disk, the collective scattering of thermal photons from dust grains by other dust grains produces polarization that is parallel to the disk minor axis (Kataoka et al. 2015; Yang et al. 2016). This pattern, indicative of scattering, has been observed in several sources (e.g., Stephens et al. 2017; Bacciotti et al. 2018; Dent et al. 2019)."***?**

"Several mechanisms have been proposed for grain alignment, including alignment to the magnetic field or radiation anisotropy through radiative alignment torques

(RATs; Tazaki et al. 2017) and mechanical alignment (Gold 1952; Kataoka et al. 2019; Yang et al. 2019)."?

""? *""*? *""*? *""*?

1.4.2 Polarization Mechanisms

[NOTE: Why/how polarization can happen, and the mechanisms that cause it]

[NOTE: In Chapter **??** *I refer to this section for a reasoning on why a circular pattern can be from radiative alignment, or some other unknown reason]*

Polarized emission of dust can be caused from many different mechanisms in disks (?). Each mechanism helps contrain different physical properties of the disk (?).

"Polarized emission at mid-infared wavelengths can occur due to absorption, emission and sometimes scattering, causing difficulty in interpreting the polarization morphology "(?)

"Observations have shown the existence of a wide range of disk polarization morphologies (?)".

The mechanism causing polarization at (sub) millimetre wavelengths in protostellar disks is unknown (??)

"Polarization patterns not consistent with pure scattering have also been observed in some protoplanetary disks at millimeter wavelengths. At 3 mm, HL Tau, DG Tau, and Haro 6-13 all exhibit an azimuthal polarization pattern that likely arises from primarily thermal emission by aligned, elongated dust grains (Stephens et al. 2017; Harrison et al. 2019)."?

1.4.3 Magnetic Alignment

"Polarized dust emission outside of disks reveal magnetic field morphology of molecular clouds" (?)

"Classically, polarization is expected from non-spherical grains aligned with the magnetic field" (?)

"Spinning dust grains in the interstellar medium are expected to align with their short axes perpendicular to the magnetic field of radiative torques" (?)

Dust Self-Scattering

$$a_{max} = \lambda/2\pi \tag{1.1}$$

"For a given observing wavelength, the polarization fraction from scattering is highly dependent on the maximum dust grain size. The polarization fraction peaks when the maximum grain size is of order $\lambda/2\pi$, where λ is the observing wavelength (Kataoka et al. 2015). Therefore, observing the scattering polarization spectrum of a source can constrain the maximum grain size"?

"If the dust grains have roughly the same size of the light you are observing, the dust grains are expected to have a large scattering opacity, therefore the continuum emission is expected to be polarized due to self-scattering (?). "

"If the grains are large sufficiently large, they have a large scattering opacity, and there is a significant amount of scattered light component in the dust continuum emission ?."

"One polarization mechanism is caused by self-scattering from dust grains of sizes

comparable with the wavelength "(?)

"In addition to the dust grain size, the disk inclination, optical depth, and dust scale height can all affect the level of polarization from scattering (e.g., Yang et al. 2017; Ohashi and Kataoka 2019; Brunngräber and Wolf 2020)."?

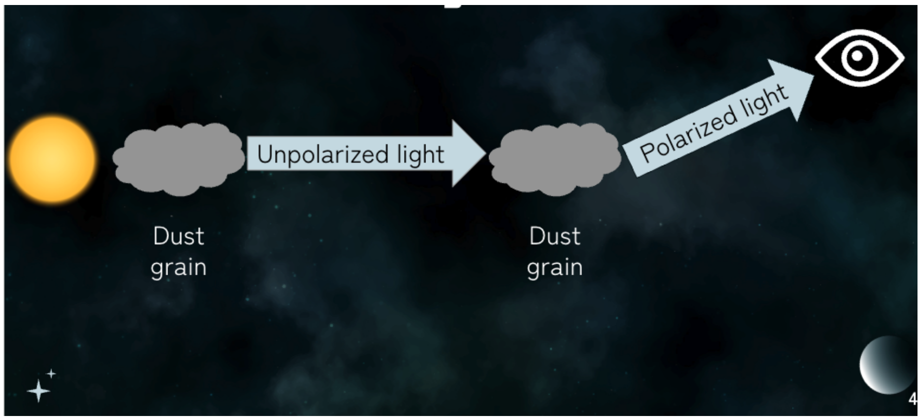

Figure 1.4: **[ADD:** caption]

Radiative Alignment

1.4.4 Polarization Morphologies

[NOTE: Pattern due to polarization]

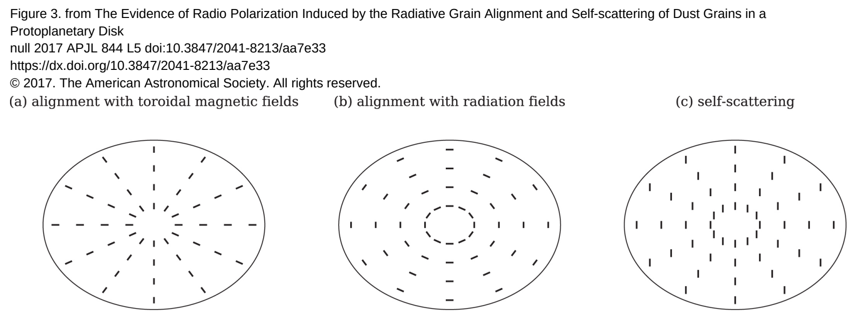

Figure 1.5: **[ADD:** caption] *[NOTE: ask permission to use] [NOTE: I will take off the top part and make it larger, better quality, etc]*

1.4.5 Why polarization is useful

1.5 Observing Polarization with ALMA

1.5.1 Why (sub)millimetre wavelength observations?

1.5.2 ALMA

The Atacama Large Millimeter/submillimeter Array (ALMA) is an international radio telescope located in the Atacama Desert in northern Chile (**?**). It has 66 antennas working together as an interferometer, with multiple array configurations with baselines ranging from 15 meters to approximately 15 km (**?**). ALMA has an observing wavelength range of 0.3 to 10 mm (**?**), allowing it to probe thermal emission from dust in protostellar disks.

Q: Add something that ALMA is high resolution and sensitity?

1.5.3 Polarization Quantities

The electromagnetic waves that ALMA measures can *[NOTE: need to find a better way to say this]* be defined in the four Stokes parameters: Stokes I represents the total intensity, Stokes Q and U describe the linear polarization, and Stokes V describes the circular polarization (**?**). In this project we only consider linear polarization and Stokes I, Q and U (i.e. no circular polarization). **Q:** no circular polarization in disks?

1.5.4 Why Multi-wavelength observations are important

As mentioned in Chapter **??**, the observed polarization morphology and inferred dominant polarization mechanism have been found to change with observing wavelength. However, the dominant polarization mechanism does not depend on wavelength alone. Different observing wavelengths are sensitive to different grain sizes and optical depths (**?**), which reveal different mechanisms. By comparing the polarization fraction and morphology at multiple wavelengths, it is possible to disentangle the polarization mechanisms and place stronger constraints on grain sizes compared to what could be achieves with observations at a single wavelength (**??**).

[NOTE: In the plot of Pw vs a max I refer to this for the importance of mw observations for grain size]

1.5.5 HL Tau

"HL Tau's polarization spectrum is particularly interesting: the disk's polarization morphology transitions from a pattern consistent with scattering at ∼870 μm to an elliptical pattern at ∼3 mm, with an intermediate pattern at ∼1.3 mm (Stephens et al. 2017). Recently, Lin et al. (2022) and Lin et al. (2024) have demonstrated that HL Tau's polarization spectrum could be explained by polarization from thermal emission and the scattering of aligned grains, also demonstrating that the polarization seen at high resolution in HL Tau at 870 μm could be explained by the same mechanisms. At longer wavelengths, thermal polarization dominates because the low optical depth makes scattering inefficient. At shorter wavelengths, scattering polarization dominates because the high optical depth makes scattering events frequent and dichroic extinction decreases the contribution from thermal polarization (Lin et al. 2022)." (**?**)

1.6 IRS 63

IRS 63 is a Class I protostar embedded in the L1709 region of the Ophiuchus molecular cloud at a distance of 144 pc (**??**). It is one of the brightest Class I protostars at (sub)millimetre wavelengths, and is surrounded by a relatively large circumstellar disk with a radius larger than 50 AU (**?**).

Since IRS 63 is relatively bright and nearby, it is an ideal target for polarization observations. Additionally, the large disk allows the polarization morphology to be studied from the inner to outer disk. Lastly, since IRS 63 is a Class I protostar, it provides the opportunity to study dust grain growth during the early stages of planet formation (**?**).

[**ADD:** previous observations and what they found]

1.7 Motivation and Objectives

1.8 Thesis Outline

../IMAGES/WRITEUP/NOTMINE/Lin2024Figure1.png

Figure 1.6: **[ADD:** caption] *[NOTE: Placeholder]*

Chapter 2

Interferometry

2.1 Introduction to Interferometers

Studying protostellar disks at millimetre wavelengths requires high-angular-resolution observations to probe their structure through the surrounding protostellar envelope. Because protostellar disks are typically located hundreds of parsecs away, they appear very small in the sky, and the angular resolution is limited by diffraction. The angular resolution, $\theta_{\rm res}$, of a single-dish radio telescope observing at wavelength λ is approximately:

$$\theta_{\rm res} \sim \lambda/D, \tag{2.1}$$

where D is the diameter of the dish (**?**). Achieving the angular resolution required to study a protostellar disk with a single-dish telescope would therefore require an impractically large dish. For example, IRS 63 is located $\sim$ 144 parsecs away, and to resolve its structure on scales $<$ 50 AU requires an angular resolution of $<$ 0.5 arcsec. At millimetre wavelengths, achieving this resolution would require a telescope around a kilometre in diameter.

To overcome this limitation, radio astronomers use interferometers, which combine observations from multiple widely separated dishes to synthesize a large dish, improving angular resolution (?). The angular resolution of an interferometer is:

$$\theta_{\mathrm{res}} \sim \lambda / B, \tag{2.2}$$

where B is the maximum distance between two telescopes, known as the baseline (?).

2.2 The Basics of Interferometry

The simplest interferometer consists of two telescopes that observe the same astronomical object. When the signals from the two antennas arrive at the same location to be combined, they either have constructive or destructive interference, producing a larger or smaller wave, respectively (?). Because the antennas are separated by some distance, radio waves from the astronomical source reach the antennas at different times, producing a geometric delay between the signals, as shown in Figure ??. The signals collected by the two antennas are combined to create an interference pattern, known as a fringe. Longer baselines make finer fringes and provide higher angular resolution, making the interferometer sensitive to smaller-scale structures. Alternatively, shorter baselines make larger fringes, causing lower angular resolution but are sensitive to larger-scale structure (?). .

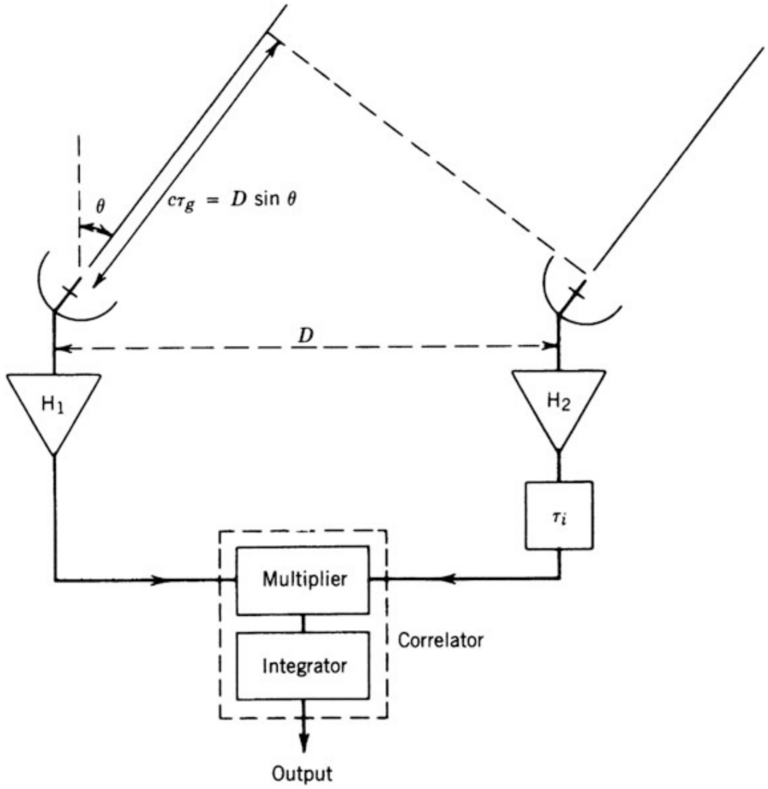

Figure 2.1: Schematic of a two-antenna ratio interferometer highlighting the geometric and instrumental delays between signals received at the two antennas. Image taken from **?** *[NOTE: Have to ask for permission, just a screenshot, can get better quality]*

2.3 UV Plane

The projected baseline between two antennas is decomposed into two coordinates (u, v) with units of the observing wavelength along the east and north directions, respectively, as projected in the direction of the target. The geometric relationship between the source direction and projected baseline coordinates is illustrated in Figure **??**. For ground-based observing (like ALMA), the object moves across the sky as the Earth rotates, so the (u, v) coverage changes with time. This is beneficial since Earth's rotation increases the (u, v) coverage without requiring additional telescopes (**?**). An example of the resulting (u, v) coverage is shown in Figure **??**

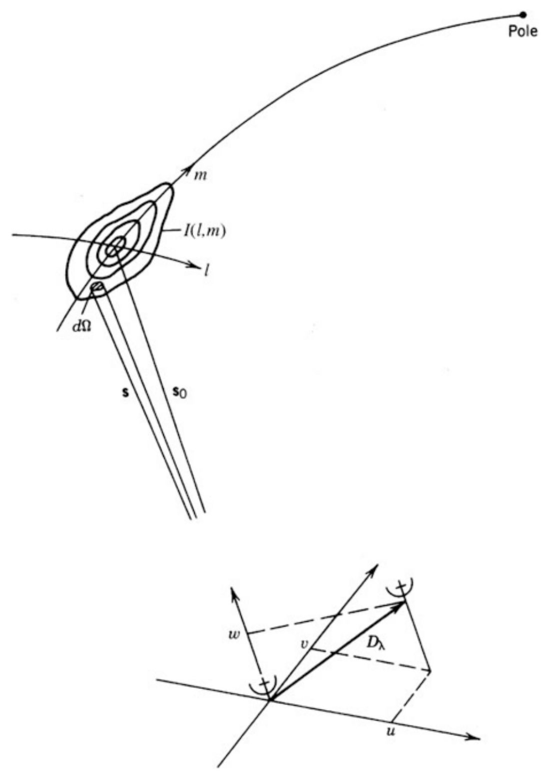

Figure 2.2: Schematic showing the geometric relationship between a source with sky coordinates (l, m) and the baseline vector of an interferometer, denoted by D_λ. The baseline is described by its components (u, v, w). Image taken from **?** *[NOTE: Have to ask for permission, just a screnshot, can get better quality]*

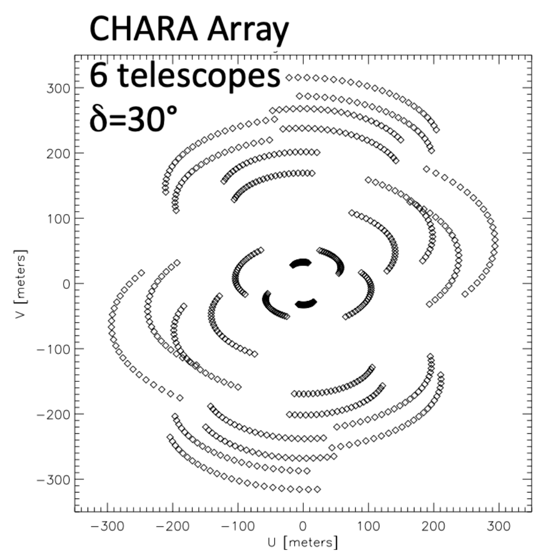

Figure 2.3: Schematic showing the (u, v) coverage. Image taken from **?** *[NOTE: Have to ask for permission, just a scren-shot, can get better quality]*

Each antenna pair measures one complex visibility, which describes the amplitude and phase of the interference pattern as a function of time and frequency (**?**). Each complex visibility is a single Fourier component of the sky brightness distribution, and together they sample its Fourier transform. The visibility function $V(u, v)$ is given by:

$$V(u, v) = \int_{-\infty}^{\infty} I(l, m) e^{-2\pi i(ul+vm)} dldm \tag{2.3}$$

where $I(l, m)$ is the intensity function, and (l, m) are the angular sky coordinates (?).

2.4 Calibration

Calibration is the process of removing the effects of instrumental and atmospheric factors from visibility data before they are combined into an image. This is achieved by observing a calibration source, an object with well-known properties. Different calibration steps require calibrators with different properties, such as brightness, size, and close proximity to the science target. Before calibration begins, contaminated data from random glitches or other bad data values are flagged and removed to make calibration as effective as possible (?).

The first step of the calibration process is bandpass calibration, which corrects for frequency-dependent instrumental effects (??). The best calibrator source for this is a very bright quasar with a flat spectrum that may not be located near the science target (?).

The next step is phase and amplitude gain calibration, which corrects time-dependent atmospheric and instrumental variations during the observation. A bright, point-like quasar close to the science target is used as a gain calibrator, as it is observed frequently throughout the observation (?).

The last step is flux calibration, which scales the relative amplitudes to the correct flux density scale. A good flux calibrator is a source with a well-known flux density, usually a Solar System object or quasar (**?**).

Polarization observations also require a polarization calibrator for polarization leakage, or the instrumental polarzation. These objects are typically unpolarized quasars that are observed for many hours at a range of elevations to measure how the instrumental polarization changes with elevation (**?**).

2.5 Imaging

2.5.1 Image Reconstruction

The inverse Fourier transform of the complex visibilities in the (u, v) plane (Equation **??**), given by the equation:

$$I(l, m) = \int_{-\infty}^{\infty} V(u, v) e^{2\pi i(ul+vm)} du dv, \tag{2.4}$$

converts the visibilities into the image domain (**?**). Since the (u, v) coverage is incomplete, this produces a 'dirty image', rather than a perfect reconstruction. The incomplete (u, v) coverage results in the dirty image being the true sky brightness distribution convolved with the point spread function of the interferometer, called the 'dirty beam' (**?**).

Weighting Schemes

Before the inverse Fourier transform is taken, each visibility in the (u, v) plane is assigned a weight, which influences the properties of the dirty beam and the resulting image (**?**). When each visibility is weighted equally, this is called *natural weighting*, which gives the highest sensitivity and therefore best signal-to-noise ratio (S/N) for detecting faint objects (**??**).

When visibilities are weighted by visibility density in the (u, v) plane, such that well-sampled regions are given less weight than less-sampled regions, this is called *uniform weighting*. Since the (u, v) plane is generally more densely sampled at shorter (u, v) distances, uniform weighting weighs the longer (u, v) distances higher. This produces a smaller synthesized beam and therefore higher angular resolution, but with lower sensitivity as a trade-off (**?**).

A 'robust' parameter was introduced in **?** for a weighting scheme called Briggs weighting. The robust parameter controls a balance between natural and uniform weighting to find an ideal trade-off between sensitivity and angular resolution (**??**). The robust parameter ranges from -2 to 2, where more negative values have weighting closer to uniform weighting, and more positive values produce weighting closer to natural weighting (**?**).

2.5.2 Deconvolution (`clean`)

Deconvolution is the process of reducing the effects of the dirty beam in the dirty image, typically using the algorithm `clean` (**??**). The `clean` algorithm iteratively builds a model of the image from delta functions or small Gaussian components. At each iteration, the brightest point of the dirty image is located, a scaled copy of the dirty beam centred on this point is subtracted, and the position and brightness are recorded as `clean` components (?). This process is repeated iteratively until only the residual noise is left. The `clean` model is then convolved with the final `clean` beam, called the synthesized beam, which is a smooth Gaussian with the same full width at half maximum (FWHM) as the dirty beam. The residual image is added to the restored image, producing the final `clean` image (?).

2.5.3 Self-Calibration

Self-calibration is an additional calibration step that can be performed after standard calibration. In standard calibration, calibration sources such as quasars are used to determine instrumental and atmospheric effects. In self-calibration, the science target itself is used as the calibrator to further correct both the phase and amplitude of the data.

The signal recieved by each antenna can be affected by time-dependent atmospheric variations, which can introduce additional calibration errors. Self-calibration is useful for reducing artifacts caused by these errors, as well as artifacts from a background source due to direction-independent calibration errors (?). The specific details on the self-calibration process used in this project are described in Chapter ??.

Chapter 3

Data

3.1 ALMA Observations

Q: What is the project ID? IRS 63 was observed in full polarization by ALMA at 2.1 mm (Band 4), 1.5 mm (Band 5), 1.3 mm (Band 6), and 0.87 mm (Band 7) between 2017 and 2022 [**ADD:** project ID]. The 1.5 mm polarization observations were originally presented in **?**, and the remaining datasets were presented in **?** as continuum only. Information about the observations is outlined in Table **??**.

3.2 Self-Calibration

In this section, the self-calibration details for this project are summarized. The data at 2.1 mm, 1.5 mm and 0.87 mm were calibrated as part of this project, and the 1.3 mm data were taken from **?**.

To clean the data, `tclean` was used in Common Astronomy Software Applications (CASA) version 5.6.1-8 (**?**). The images were cleaned using the multi-scale multi-frequency deconvolution algorithm (MS-MFS, `deconvolver = 'mtmfs'`). Different

Table 3.1: ALMA observation and self-calibration information for IRS 63 at each wavelength. *[NOTE: **?** also has 2017-05-13 and 2017-05-15 for observing dates for Band 6.] [NOTE: **?** has number of antenna as 46, 45, 57, 42. also time of source as 8.06 for Band 6] [NOTE: **?** has time of source as 8.06 for Band 6 and Sarah commented maybe 7.3 min]*

Property	2.1 mm	1.5 mm	1.3 mm	0.87 mm
Observation Date	2019-09-01	2022-08-10	2017-05-20	2022-08-21
	2019-09-02	...	2017-07-11	...
	...	...	2017-07-14	...
ν (GHz)	145	203	233	344
Phase Calibrator	J1633-2557	J1617-2537	J1625-2527	J1626-2951
Flux/Bandpass Calibrator	J1517-2422	J1517-2422	J1517-2422	J1517-2422
Polarization Calibrator	J1751+0939	J1751+0939	J1549+0237	J1751+0939
Time on Source (min)	37.7	35.62	≈ 7	30.81
Min. Baseline (m)	38	15	15	15
Max. Baseline (m)	3638	1302	2647.3	1302
# Antenna	47	45	42-46	45

values of `nterms` were explored, and no significant change between `nterms = 1` and `nterms = 2` was found. The images presented in this thesis use `nterms = 2` for 2.1 mm and 0.87 mm, `nterms = 1` for 1.5 mm and 1.3 mm. The Briggs weighting scheme was used with `robust = 0.5` for all wavelengths except the 1.5 mm data, where `robust = -1` was used to better match the resolution of the other datasets (see Chapter **??** for more details).

The Stokes I images at each wavelength were interactively cleaned during the self-calibration process. We began with a shallow phase-only self-calibration, and progressively cleaned deeper each time for a total of three rounds. The solution intervals were infinity, 30.3 seconds (15 integrations) and 20.2 seconds (10 integrations). Then, two rounds of phase and amplitude self-calibration were performed.

The solution intervals were infinity and 100 seconds (50 integrations). After each round of self-calibration, the phase and amplitude gain solutions were examined to ensure the solutions were reasonable before being applied to the data.

Additionally, the Stokes Q and Stokes U data were cleaned using a non-interactive phase calibration using the final mask from the amplitude calibration at its respective wavelength. The same Briggs `robust` value used for Stokes I was used for Stokes Q and U, with `nterms = 1`. *[NOTE: Checked this for Band 4 and 7. Can check 5 and 7 when I can access my data again.]*

Figure **??** shows an example of IRS 63 at 2.1 mm before and after self-calibration. The background noise is reduced after self-calibration, and the S/N improves from 660 to 2655. Similar improvements were seen for the other wavelengths.

Table **??** lists the resulting image resolution after self-calibration, including the synthesized beam parameters and clean details. The table also includes the disk centre position in Right Ascension (RA) and Declination (Dec), as well as the disk position angle found by fitting a Gaussian to the disk using `imfit` in CASA. The image sensitivities are also provided and were determined using the method described in Chapter **??**.

3.3 Debias Correction

The observed polarized intensity, $\mathcal{P}_{\rm obs}$, is calculated from Stokes Q (Q) and Stokes U (U) using the equation:

$$\mathcal{P}_{\rm obs} = \sqrt{Q^2 + U^2}. \tag{3.1}$$

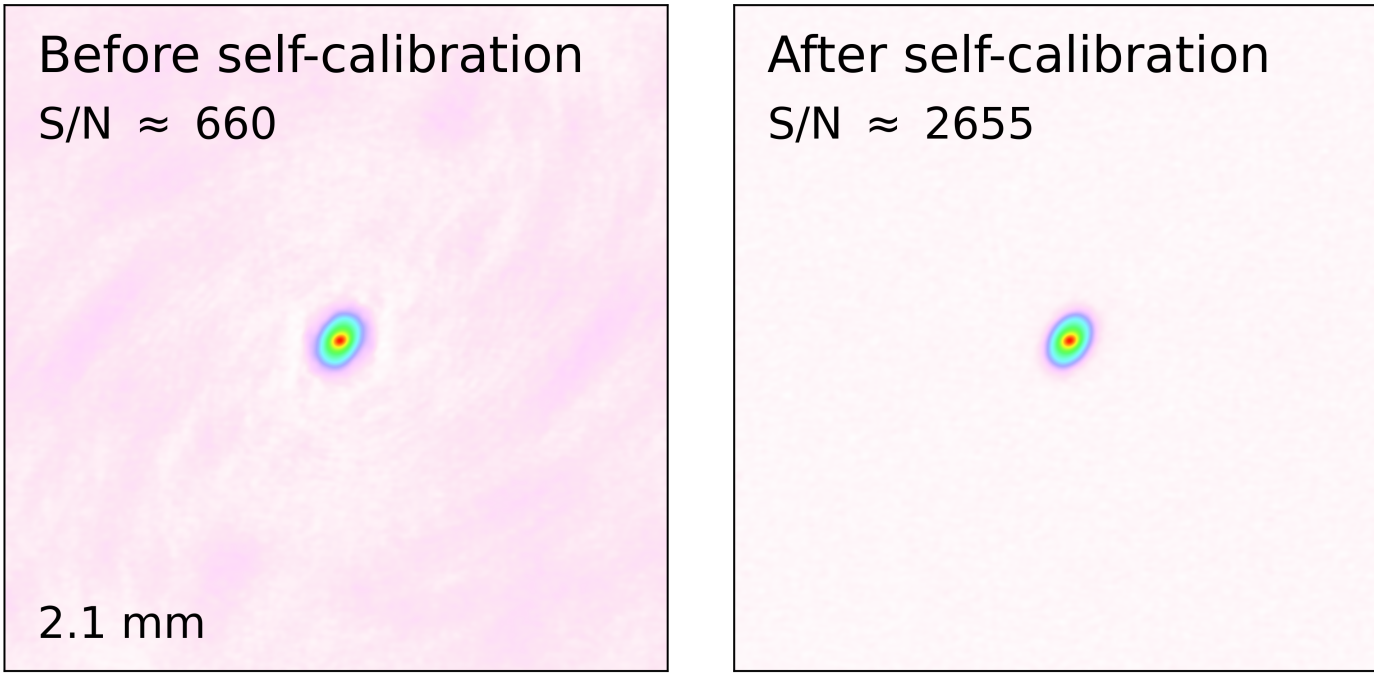

Figure 3.1: Images before (left) and after (right) self-calibration at 2.1 mm. The images demonstrate how background noise is reduced during the self-calibration process, improving the peak S/N.

Since Stokes Q and U are squared, the equation always yields a positive value. This introduces a positive bias when either Stokes Q or U are dominated by noise, since noise can cause non-zero values even when there is no polarized emission. To account for this bias, a debiased polarized intensity is calculated.

The polarized intensity was debiased using the method explained in **?**, originally from **?**. Variables used in these equations and further into this section are summarized in Table **??**. The following equations in this Chapter are from **?**, unless specified otherwise.

This method calculates the debiased polarized intensity, $\mathcal{P}_I$, with a corresponding

Table 3.2: Summary of the properties of the cleaned images of IRS 63 at each wavelength. **Q:** Can I give + 83 as - 97?

Property	2.1 mm	1.5 mm	1.3 mm	0.87 mm
R.A. (h, m, s)	16:31:35.656	16:31:35.656	16:31:35.657	16:31:35.656
Decl. (J2000)	-24:01:29.992	-24:01:30.086	-24:01:29.935	-24:01:30.089
Beam ($'' \times ''$)	0.18 × 0.13	0.23 × 0.18	0.27 × 0.20	0.28 × 0.17
Beam PA (°)	-81.0	+83.0	-68.6	-81.2
Disk PA (°)	145.9	148.39	148.0	139.7
`nterms`	2	1	1	2
`robust`	0.5	-1	0.5	0.5
σ_I	13.75	42.12	71.01	35.17
σ_Q	9.65	40.50	27.00	26.25
σ_U	10.12	40.80	27.00	24.35

Note: σ_I, σ_Q, σ_U are constant values for 2.1 mm, 1.5 mm, and 0.87 mm, and are reported at maximum Stokes I value for 1.3 mm. These errors are reported in units of μJy beam^{-1}.

Table 3.3: Summary of Stokes and polarization parameters, their associated error symbols, and units.

Parameter	**Symbol**	**Error Symbol**	**Units**
Stokes I	I	σ_I	mJy beam^{-1}
Stokes Q	Q	σ_Q	mJy beam^{-1}
Stokes U	U	σ_U	mJy beam^{-1}
Observed (Biased) Polarized Intensity	$\mathcal{P}_{\rm obs}$	N/A	mJy beam^{-1}
Debiased Polarized Intensity	$\mathcal{P}_I$	$\sigma_{\mathcal{P}I}$	mJy beam^{-1}
Polarization Fraction (debiased)	$\mathcal{P}_F$	$\sigma_{\mathcal{P}F}$	unitless
Polarization Angle (debiased)	θ	σ_θ	°

error of $\sigma_{\mathcal{P}I}$, with the equation:

$$\mathrm{PDF}(\mathcal{P}_I|\mathcal{P}_{\rm obs}, \sigma_{\mathcal{P}I}) = \frac{\mathcal{P}_{\rm obs}}{\sigma^2_{\mathcal{P}I}} I_0(x) \exp\left[-\frac{(\mathcal{P}^2_{\rm obs} + \mathcal{P}^2_I)}{2\sigma^2_{\mathcal{P}I}}\right], \tag{3.2}$$

where PDF is the probability density function. Here, $I_0(x)$ is the Bessel function for the Bessel parameter $x = \mathcal{P}_{\rm obs}\mathcal{P}_I/\sigma^2_{\mathcal{P}I}$. The most likely $\mathcal{P}_I$ corresponds to the peak of the PDF in Equation ??. The debiased value is calculated by minimizing the negative value of the PDF in Equation ?? using `scipy.optimize.minimize_scalar`.

Only the pixels with a low to moderate S/N (< 9) were debiased with the method described above. The pixels with S/N> 9 do not need to be treated the same way because their emission is much stronger than the noise. These pixels are debiased using the standard maximum likelihood calculation method from ?:

$$\mathcal{P}_{I,\rm bright} = \sqrt{Q^2 + U^2 - \sigma^2_{\mathcal{P}I}}. \tag{3.3}$$

The corresponding error of the debiased polarized intensity is calculated by:

$$\sigma_{\mathcal{P}I} = \left[\frac{(Q\sigma_Q)^2 + (U\sigma_U)^2}{Q^2 + U^2}\right]^{1/2}. \tag{3.4}$$

3.4 Additional Polarization Quantities & Errors

3.4.1 Errors for Stokes Parameters

For the data calibrated in this project (2.1 mm, 1.5 mm, 0.87 mm), the errors for Stokes I, Stokes Q and Stokes U were estimated by averaging the maximum standard deviation in four different regions far from the bright source at the centre. This average value was applied as a constant error at every pixel in the respective error maps. The values are calculated in mJy but are presented in Table ?? in units of μJy.

3.4.2 Polarization Fraction

Using the debiased polarized intensity, the (debiased) polarization fraction can be calculated by:

$$\mathcal{P}_F = \mathcal{P}_I / I, \tag{3.5}$$

where I is Stokes I intensity. The uncertainty on polarization fraction is:

$$\sigma_{\mathcal{P}F} = \mathcal{P}_F \left[\left(\frac{\sigma_{\mathcal{P}I}}{\mathcal{P}_I} \right)^2 + \left(\frac{\sigma_I}{I} \right)^2 \right]^{1/2}. \tag{3.6}$$

3.4.3 Polarization Angle

The polarization position angle, θ is calculated by:

$$\theta = \frac{1}{2} \tan^{-1} \frac{U}{Q}, \tag{3.7}$$

with an uncertainty calculated by:

$$\sigma_\theta = \frac{1}{2} \frac{\sqrt{(U\sigma_Q)^2 + (Q\sigma_U)^2}}{Q^2 + U^2}. \tag{3.8}$$

Chapter 4

Observational Analysis

4.1 Making Polarization Maps

The polarization morphology of a disk is represented by polarization vectors calculated from Stokes Q and U, which define the polarization angle (Equation **??**). From these morphologies, the dominant polarization mechanism can be inferred, as different mechanisms produce different polarization patterns (Chapter **??**). By comparing the polarization morphologies at multiple wavelengths, we can examine how the dominant polarization mechanism changes with optical depth. We first calculated the polarization vector and then created models to fit the observed morphologies and determine the dominant mechanisms.

4.1.1 Polarization Vector Sampling

The Stokes I maps produced using `tclean` have a pixel scale that oversamples the synthesized beam to improve image reconstruction (**?**). Because of this, plotting a polarization vector at each pixel would oversample the polarization morphology.

Instead, vectors are plotted with a spacing of approximately half of the synthesized beam size, following the Nyquist-Shannon sampling theorem (**?**).

The effective beam size was calculated using:

$$\text{Beam Size} = \sqrt{BMIN \cdot BMAJ}, \tag{4.1}$$

where $BMIN$ and $BMAJ$ are the synthesized beam minor and major axes, respectively, listed in Table **??** in units of arcseconds. The beam dimensions were converted from arcseconds to pixels using the image pixel size of $0.018''\,\text{pixel}^{-1}$. The polarization vector spacing was then set to be half the beam size (Equation **??**), and rounded to the nearest integer number of pixels.

4.1.2 Reliability Criteria

In addition to the sampling spacing described above, a polarization vector was only plotted if it satisfied the following criteria:

1. $I/\sigma_I > 3$,
2. $\mathcal{P}_I/\sigma_{\mathcal{P}I} > 4$ (2.1 mm, 0.87 mm) or $\mathcal{P}_I/\sigma_{\mathcal{P}I} > 3$ (1.5 mm, 1.3 mm),
3. $\sigma_\theta < 10°$.

These criteria were adapted from the selection criteria in **?**. The threshold of $\mathcal{P}_I/\sigma_{\mathcal{P}I} > 3$ was used for 1.5 mm and 1.3 mm because of their sensitivities: threshold of 4 excluded many additional polarization vectors.

4.2 Observed Polarization Maps

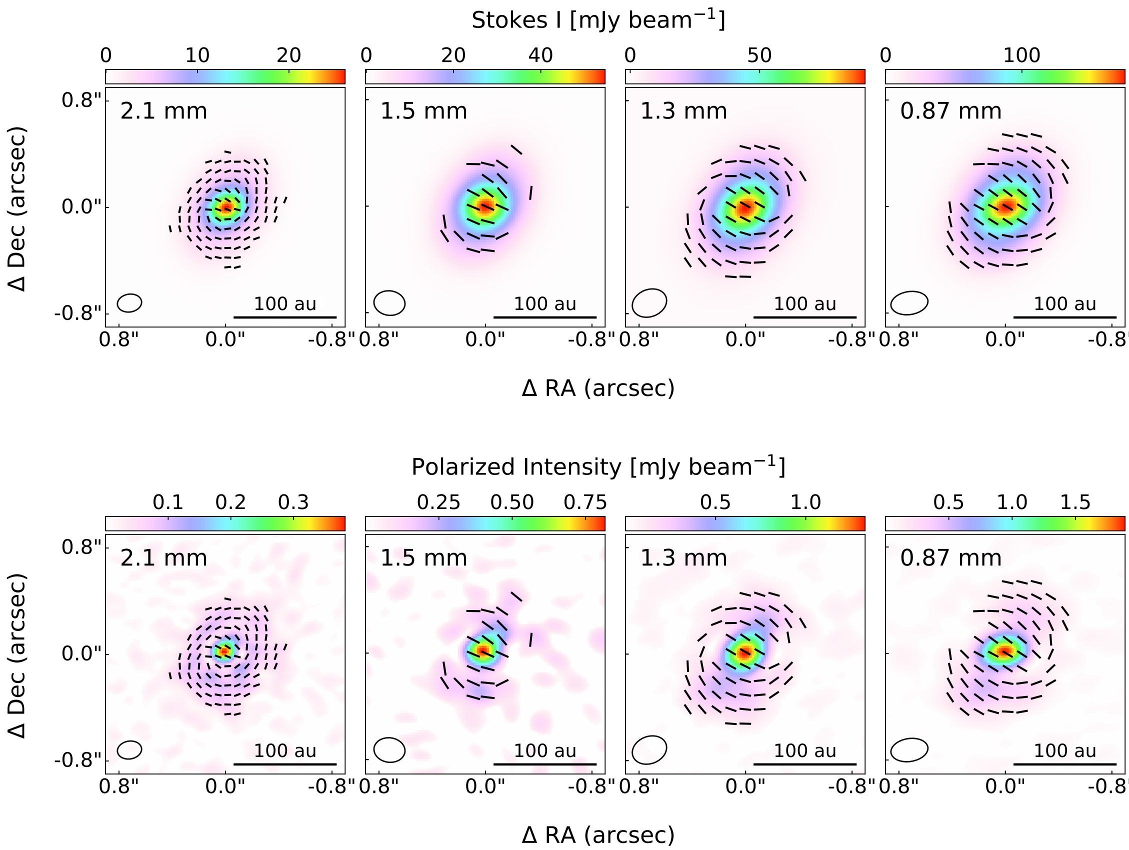

Figure 4.1: Grid of images at each wavelength showing (top) Stokes I emission and (bottom) debiased polarized intensity. Polarization vectors are overlaid as black lines of equal length, showing the change in morphology with wavelength. Vectors are shown only for pixels that satisfy the criteria described in Chapter **??**. The beam is shown in the lower left corner, and a reference scale of 100 AU is shown in the top right corner. **Q:** scale vectors by POLF?

In Figure **??**, we present the cleaned Stokes I and debiased polarized intensity images of IRS 63 at 2.1 mm, 1.5 mm, 1.3 mm, and 0.87 mm, overlaid with the polarization vectors that make up the observed polarization morphology. As the observing wavelength decreases, there is a clear transition in polarization morphology.

At 2.1 mm, the polarization vectors are mostly azimuthal, forming a circular pattern around the centre of the disk. This is consistent with the pattern expected from radiative alignment (or some other mechanism, Chapter **??**), shown in Figure **??**(b). However, near the centre of the disk, there is a small region where the vectors are aligned with the minor axis of the disk. This pattern is consistent with dust self-scattering, as shown in Figure **??**(c).

As the wavelength decreases from 2.1 mm to 0.87 mm, the centre region where vectors are aligned with the minor axis becomes larger, while the azimuthal pattern resides only on the outer part of the disk. The patterns at 1.5 mm and 1.3 mm are more of a transition, but at 0.87 mm the uniform alignment consistent with dust self-scattering is clearly dominating.

A similar wavelength-dependent change in polarization morphology has been presented for HL Tau by **?**, shown in Figure **??** for the same observing wavelengths as this project. For both objects, the polarization morphology transitions from being mainly azimuthal at longer wavelengths to becoming more uniformly aligned with the minor axis at shorter wavelengths. However, there are some noticeable differences. In HL Tau, at longer wavelengths, the vectors close to the centre of the disk still have an azimuthal pattern and do not have the uniform alignment seen in the centre of IRS 63. Additionally, at longer wavelengths, the vectors show a more uniform pattern across the entire disk, whereas IRS 63 has an apparent azimuthal contribution in the outer part of the disk at the same wavelength. These similarities and differences suggest that both disks have the same transition in the dominant polarization mechanism, but their morphologies are not identical at each wavelength.

4.3 Intensity Profiles

To study the structure of IRS 63, intensity profiles were extracted along the minor axis in both Stokes I and debiased polarized intensity at each wavelength. The locations of the slices are shown in Figure **??**.

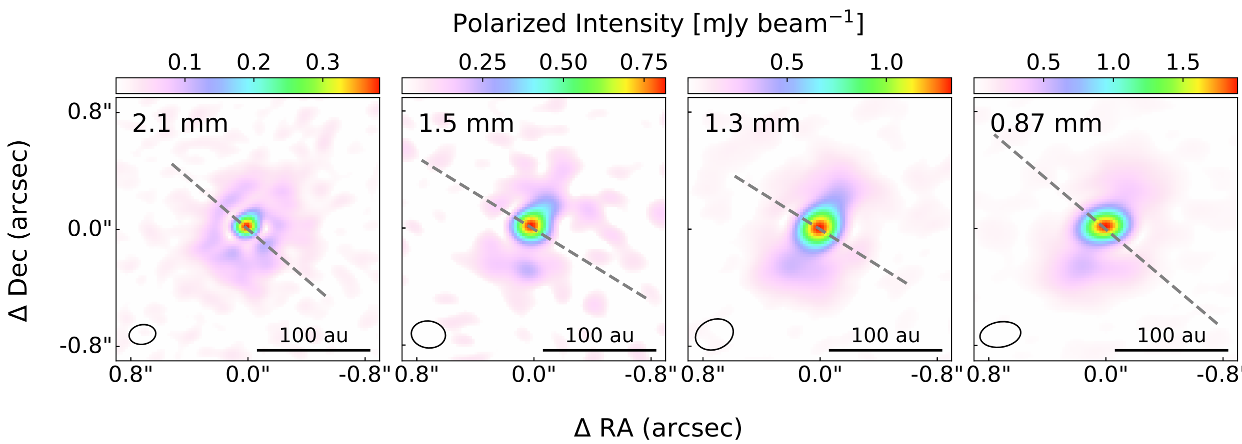

Figure 4.2: Locations of the minor-axis slices extracted from the images at each wavelength. The same slice was used for both Stokes I and debiased polarized intensity.

The slices in Stokes I and polarized intensity for each wavelength are presented in Figure **??**. The synthesized beam for each wavelength is overplotted for comparison. The beam is plotted as a one-dimensional Gaussian, with a standard deviation calculated from the beam full width at half maximum (FWHM) along the minor axis using:

$$\sigma = \frac{\mathrm{FWHM}}{2\sqrt{2\ln 2}}. \tag{4.2}$$

The Gaussian was normalized to 1 and scaled by the peak intensity of each slice.

The Stokes I profiles (top row of Figure **??**) show a single peak at the centre of the disk at all four wavelengths. This is as expected, as the protostar at the centre of the disk should be producing the most emission. However, in the debiased polarized intensity profiles (bottom row of Figure **??**), we do not see the same trend. The profiles are consistently offset from the centre position at all four wavelengths. The offset is larger than the synthesized beam size, indicating that it is spatially resolved.

A similar signature, where the polarized intensity is brighter on the near side of the disk, was shown in **?**. This indicates that the disk is optically thick and flared, meaning that the disk scale height increases with radius (Figure **??**). The angular resolution of the data is not high enough to further explore this signature or resolve smaller structures in the disk.

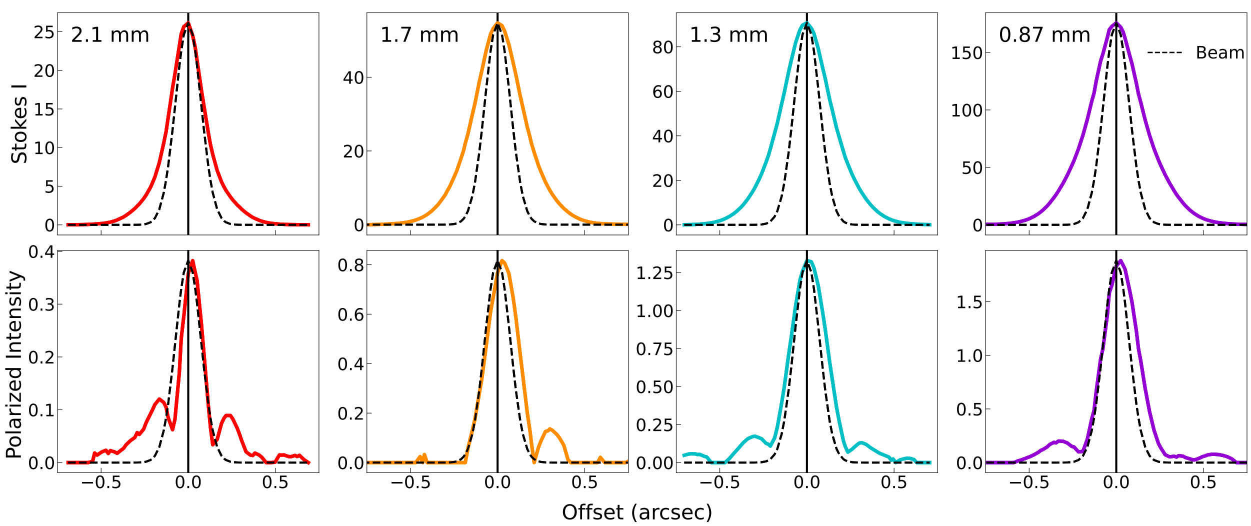

Figure 4.3: Plot of (top) Stokes I and (bottom) polarized intensity offset as a function of the offset from the centre position of the disk, both in units of mJy beam^{-1}. The centre position is shown as a solid black line, with the corresponding values found in Table **??**. The slices used for these profiles are shown in Figure **??**.

Chapter 5

Modelling Polarization

The scattering models for spherical and irregular dust grains are presented, with various plots reproduced from the literature to ensure correct implementation. The results of the scale factor plots are used to constrain the maximum dust grain size.

5.1 Geometric Modelling of Polarization Pattern

Once the polarization vector maps were created, the observed polarization morphologies were modelled using two geometric models. The models were created to reproduce the observed polarization morphologies, including azimuthal patterns (Figure **??**a), polarization vectors that uniformly align with the minor axis (Figure **??**b), and intermediate morphologies. Since dust self-scattering is expected to produce a uniform pattern, the model allows us to identify the regions of the disk where scattering-induced polarization may dominate.

A more physical model could include mechanisms such as radiative transfer, but purely geometric models are sufficient for this project. The goal is to identify the regions of the disk where the signatures of each polarization mechanism dominate, rather than to model the physical mechanisms that produce the observed patterns. The two geometric models are described below.

5.1.1 Ratio Model

The first geometric model tested is the *ratio model*. The main idea behind this model is to reproduce the observed polarization morphologies, ranging from purely aligned with the minor axis of the disk to purely azimuthal, along with intermediate cases. This was achieved by creating a grid of models weighted between these two extreme cases.

The first extreme case is a 100% uniform pattern, where every vector is aligned with the minor axis of the disk (90° from the disk position angle, Table **??**). This is shown in the left panel of Figure **??**. The second extreme case is a 100% azimuthal pattern, where the polarization vectors follow a circular pattern around the centre of the disk. The centre point of the disk at each wavelength can be found in Table **??**. For each pixel (x, y), the offset from the disk centre (x_c, y_c) is calculated as:

$$\begin{aligned} \delta x &= x - x_c, \\ \delta y &= y - y_c. \end{aligned} \tag{5.1}$$

These offsets are then used to calculate the azimuthal polarization angle using `numpy.arctan2`. This is shown in the right panel of Figure **??**.

To create the intermediate cases, models were generated in steps of 10% and are denoted as $N\text{Uni} + (100 - N)\text{Azi}$, where $N = 0, 10 \ldots, 100$. Here, Uni and Azi represent the pure uniform and pure azimuthal cases, respectively, and N gives the percentage contribution of the uniform case.

Since the polarization angles cannot be directly combined, the Stokes Q and Stokes U for the extreme cases ($100\text{Uni} + 0\text{Azi}$ and $0\text{Uni} + 100\text{Azi}$) were calculated from the polarization angle grids using:

$$\begin{aligned} Q_x &= pI\cos(2\theta_x), \\ U_x &= pI\sin(2\theta_x), \end{aligned} \tag{5.2}$$

following **?**, where $x = 100$ corresponds to the pure uniform model and $x = 0$ corresponds to the pure azimuthal model. The polarization fraction p was fixed at 0.02 based on observations presented in **?**. This value is expected to be reasonable for the uniform polarization morphology, but the polarization fraction may differ for the azimuthal morphology, potentially biasing the azimuthal case low.

The intermediate Stokes Q and U maps were then obtained by combining the two extreme models using:

$$\begin{aligned} Q_N &= \frac{N}{100}\,Q_{100} + \frac{100-N}{100}\,Q_0, \\ U_N &= \frac{N}{100}\,U_{100} + \frac{100-N}{100}\,U_0. \end{aligned} \tag{5.3}$$

The resulting Q_N and U_N maps were used to calculate θ_N at each pixel using Equation **??**. The polarization vectors were generated using the methods in Chapter **??**. This process was repeated for each ratio combination. For example, $N = 50$ corresponds to the $50\text{Uni} + 50\text{Azi}$ model, shown in the middle panel of Figure **??**.

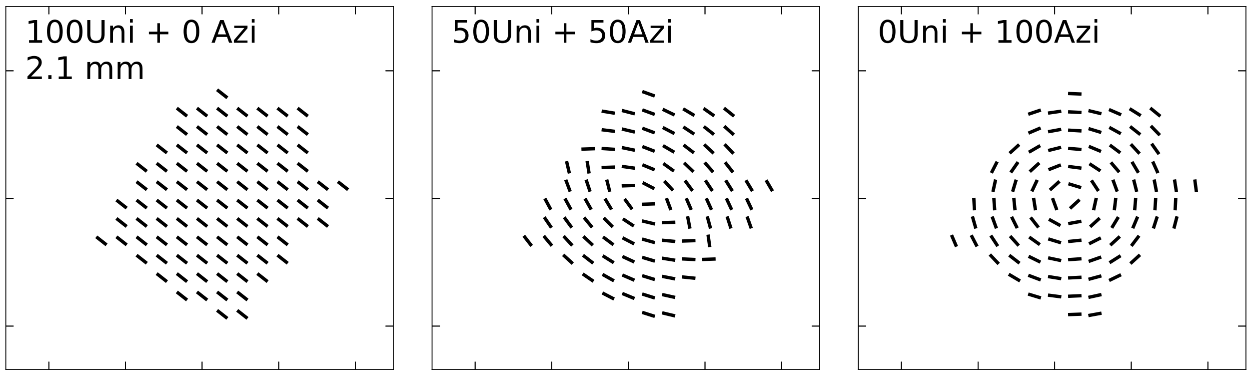

Figure 5.1: Example of the $N = 100, N = 50$ and $N = 0$ cases of the ratio model for 2.1 mm.

5.1.2 Gaussian Model

The second geometric model tested is the *Gaussian model*. The aim of this model was to reproduce a polarization morphology that transitions from being aligned with the minor axis of the disk at the centre to becoming more azimuthal towards the outer parts of the disk. This is based on the expectation that the emission from the inner region is optically thick (**?**), with optical thickness decreasing towards the outer edge of the disk.

To achieve this, we used a rotated 2-dimensional flat-topped Gaussian model with three free parameters: ϕ, BMIN and BMAJ, summarized in Table **??**. The tested values of BMIN and BMAJ control the widths of the Gaussian model, with each treated as the FWHM and converted to standard deviations using Equation **??**. Additionally, a condition of BMIN $\leqq$ BMAJ was required to ensure the minor axis was smaller than the major axis.

The Gaussian was rotated to align with the major axis of the disk, using the

Table 5.1: Summary of parameters and tested values in the Gaussian model.

Property	Meaning	Tested Values
ϕ	super-Gaussian exponent	2, 3, 4, 5, 6, 7
BMIN (pixels)	minor axis of Gaussian	10, 20, 30, 40, 50
BMAJ (pixels)	major axis of Gaussian	10, 20, 30, 40, 50

position angle of the disk given in Table **??**). The coordinates (x, y) were shifted with respect to the centre of the disk (x_c, y_c) and then rotated:

$$\begin{pmatrix} X_M \\ Y_m \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x - x_c \\ y - y_c \end{pmatrix}. \tag{5.4}$$

The Gaussian was calculated using the equation:

$$W_{\text{Uniform}} = I_0 \exp\left[-\frac{1}{2}\left(\left|\frac{X_M}{\sigma_M}\right|^\phi + \left|\frac{Y_m}{\sigma_m}\right|^\phi\right)\right], \tag{5.5}$$

where I_0, the peak value, was calculated using the equation:

$$I_0 = \frac{1}{\sigma_m \sigma_M \sqrt{2\pi}}. \tag{5.6}$$

In Equation **??**, $\phi = 2$ corresponds to a standard 2D Gaussian, while $\phi > 2$ produced a flat-topped Gaussian. The Gaussian was normalized to a maximum of 1 so that it could be used as a pixel weight. This produced a map of weights ranging from 0 to 1, where W_{Uniform} sets the weight for the uniform morphology, and 1 - W_{Uniform} sets the weight for the azimuthal morphology. With this approach, values of W_{Uniform} close to 1 favour uniform polarization and values close to 0 favour azimuthal polarization.

These two maps act as our weights, rather than a constant like in the ratio model. The weighted Stokes Q and U were calculated by:

$$\begin{aligned} Q_{\text{Gaussian}} &= W_{\text{Uniform}}\, Q_{100} + (1 - W_{\text{Uniform}})\, Q_0, \\ U_{\text{Gaussian}} &= W_{\text{Uniform}}\, U_{100} + (1 - W_{\text{Uniform}})\, U_0, \end{aligned} \tag{5.7}$$

As in the ratio model, the maps of Q_{Gaussian} and U_{Gaussian} were used to calculate the position angle (Equation **??**).

Two examples of the Gaussian weight map, with polarization vectors overlaid, are shown in Figure **??** for selected values of ϕ, BMIN, and BMAJ.

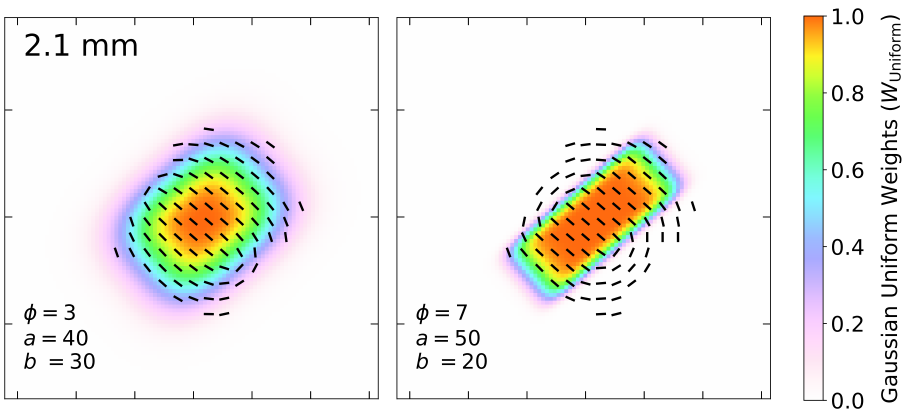

Figure 5.2: Two examples of the Gaussian weights for selected values of ϕ, BMAJ and BMIN.

*Comment: Perhaps give the equation of the flat-topped Gaussian so that the reader knows what this is. You can also mention that phi=2 is a standard Gaussian. *[NOTE: Leaving this comment here in case I did not address it properly, but I have a sentence saying "In Equation* **??***, when $\phi = 2$ the equation returns a standard 2D Gaussian, and values of $\phi > 2$ create a flat-topped Gaussian.", and Equation* **??** *shows the* *flat-topped Gaussian already. Maybe I should reorder things or provide a general flat-topped Gaussian equation?]*

5.1.3 Comparison of Geometric Models

For both the ratio model and the Gaussian model, many combinations were tested. To quantify the *goodness-of-fit* of each model, a chi-squared value was calculated.

First, the difference between the observed and model-predicted polarization angle needed to be calculated carefully. Polarization angles can range from 0–180° because polarization measures orientation, rather than direction, meaning that angles that are 180° apart have the same polarization vector orientation. As a result, two polarization angles that are physically very similar can have a large numerical difference. For example, polarization angles of 1° and 179° only differ by 2°, but directly taking their difference would give 178°. To account for this, the angular difference is defined by the minimum angular separation between the two angles:

$$\Delta\theta = \min(|\theta_{\text{obs}} - \theta_{\text{model}}|, 180^\circ - |\theta_{\text{obs}} - \theta_{\text{model}}|), \tag{5.8}$$

where θ_{obs} is the observed polarization angle and θ_{model} is the polarization angle predicted by either the ratio or Gaussian model. The angular differences were then used to calculate the chi-squared value:

$$\chi^2 = \sum \left(\frac{\Delta\theta_i}{\sigma_\theta} \right)^2, \tag{5.9}$$

where the angular distance is summed over all sampled polarization vectors.

The value of χ^2 was then converted to reduced chi-squared by:

$$\chi^2_{reduced} = \frac{1}{n-k} \times \chi^2, \tag{5.10}$$

where n is the number of sampled polarization vectors and k is the degrees of freedom. For the ratio model, there is one degree of freedom, and for the Gaussian method, there are three (ϕ, BMIN, and BMAJ).

Ratio Model

The results of the tested ratio models are presented in Table **??**, with the best fit at each wavelength shown in bold. The best fit at each wavelength is shown in Figure **??**, with the observed vectors in black and the model vectors in red.

Taking 1.5 mm as an outlier (see Chapter **??**), the best-fit ratio models show that, as wavelength decreases, the observations are better matched with a higher uniform contribution. This result is consistent with the observations discussed in Chapter **??**, which show a greater uniform contribution with decreasing wavelength.

Gaussian Model

The five best-fitting Gaussian models (out of 150 tested) are presented in Table **??**. The best fit at each wavelength is highlighted in Figure **??**, with the observed vectors in black and the model vectors in red. Additionally, the Gaussian (W_{Uniform}, Equation **??**) for each best-fit are shown in Figure **??**.

From the results in Table **??**, BMAJ and BMIN remain the same among the top five models at each wavelength, indicating that these values provide a good fit among

Table 5.2: Results of ratio model testing for each wavelength, with the goodness-of-fit quantified using the reduced χ^2 value calculated using Equation **??**. The best fit at each wavelength is in bold.

Ratio Model	2.1 mm	1.5 mm	1.3 mm	0.87 mm
100Uni + 0Azi	181.67	**22.58**	37.39	**79.12**
90Uni + 10Azi	171.40	24.22	33.93	88.59
80Uni + 20Azi	158.01	27.77	**33.36**	108.65
70Uni + 30Azi	140.74	34.28	37.99	144.84
60Uni + 40Azi	116.11	45.10	51.96	206.76
50Uni + 50 Azi	**88.21**	63.81	84.48	322.01
40Uni + 60Azi	124.99	91.17	140.41	498.43
30Uni + 70 Azi	131.03	116.35	191.20	647.50
20Uni + 80Azi	133.70	138.36	233.90	776.00
10Uni + 90Azi	136.42	157.27	270.41	886.40
0Uni + 100Azi	139.41	173.26	302.34	979.97

the tested values. The parameter that varies is the Φ parameter, which controls how flat the Gaussian is. A higher Φ value produces a flatter Gaussian and a faster transition from uniform to azimuthal polarization (see Figure **??**). In the best-fit model at each wavelength, Φ is greater than 2, producing a flat-topped Gaussian rather than a standard Gaussian. This can also be seen in Figure **??**, which shows that the area of uniform alignment increases as wavelength decreases. This indicates that uniform alignment, which is consistent with dust self-scattering, contributes more at shorter wavelengths, which is what we expect from Chapter **??** and the results from the ratio model.

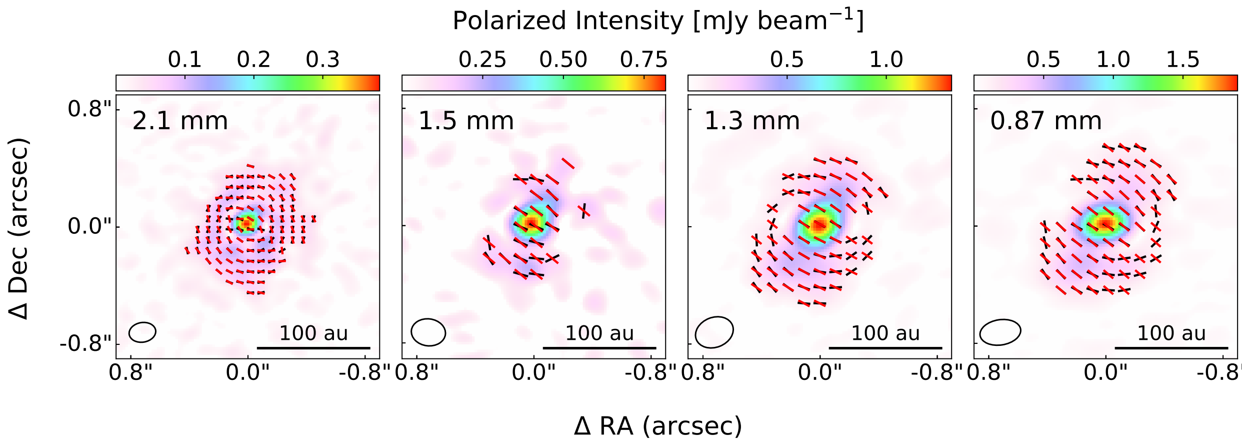

Figure 5.3: Figure **??**, overplotted with the polarization vectors in red from the best-fit ratio model at each wavelength. The corresponding best-fit ratio models are listed in Table **??**.

Ratio vs Gaussian Models

The reduced χ^2 values and visual comparisons of the ratio and Gaussian models show that the Gaussian model provides a better overall fit.

Figure **??** show that, while the ratio model matches many of the observed vectors at each wavelength, it is a poor match for others. This is a limitation of applying the same polarization pattern across the whole disk. As discussed above, the polarization pattern changes with distance from the centre of the disk. Therefore, while the ratio model provides a good first test, it is an oversimplification to accurately model the pattern throughout the entire disk.

Figure **??** shows that, although not all vectors match perfectly, many are accurately reproduced by the Gaussian model. At 1.3 mm and 0.87 mm, the model does not match the vectors along the lobes, which could be related to depolarization. **Q:** go into more detail about this? Overall, the Gaussian model is the better model but

Table 5.3: Results of the top 5 best fits (in descending order) at each wavelength from the Gaussian model. χ^2 values were calculated using Equation **??**. The best fit at each wavelength is in bold.

Wavelength	ϕ	**BMAJ**	**BMIN**	χ^2
2.1 mm	**7**	**20**	**10**	**26.70**
	6	20	10	26.79
	5	20	10	26.95
	4	20	10	27.30
	3	20	10	28.04
1.5 mm	**4**	**30**	**20**	**13.88**
	5	30	20	13.92
	6	30	20	14.09
	3	30	20	14.16
	7	30	20	14.30
1.3 mm	**4**	**30**	**30**	**8.08**
	5	30	30	8.09
	6	30	30	8.19
	7	30	30	8.33
	3	30	30	8.41
0.87 mm	**6**	**40**	**30**	**38.56**
	5	40	30	38.59
	7	40	30	38.69
	6	30	30	38.86
	7	30	30	38.88

still has limitations, as it is only geometric.

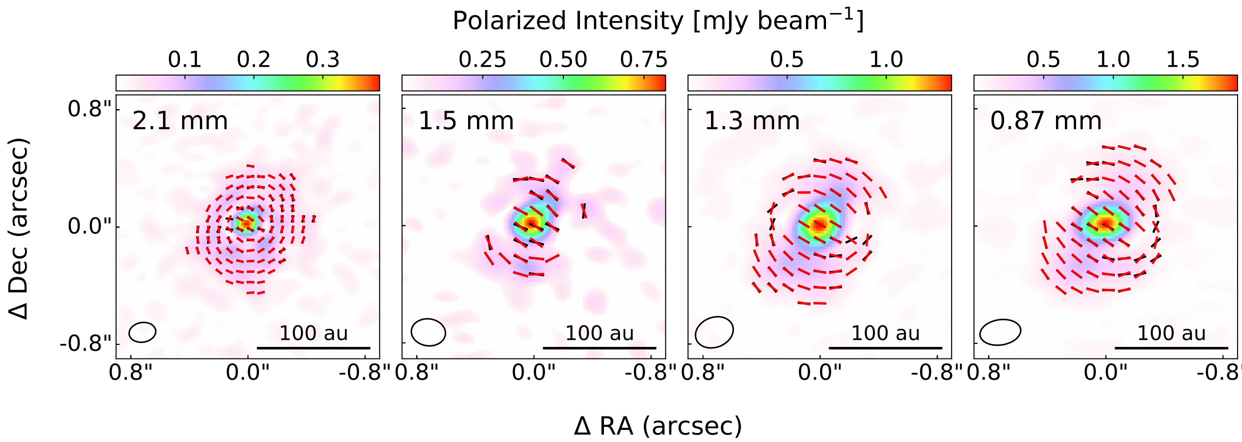

Figure 5.4: Same as Figure **??**, but for the best-fit Gaussian models. The corresponding best-fit Gaussian models are listed Table **??**.

5.2 Dust Scattering Models

Modelling the polarization from scattering across the multi-wavelength data allows us to constrain the dust grain size using a method adapted from **?**, which is detailed below. Two grain geometries were explored: spherical grains and irregular grains. Both use the same dust composition and grain-size distribution (Chapter **??**), but have different optical properties due to their different shapes. Spherical dust grains are more commonly used because their scattering and absorption properties are less computationally expensive. Irregular dust grains are more realistic (**?**), but require more computational power.

The geometric modelling of the observations in Chapter **??** was used to identify the region of the disk at each wavelength where the observed polarization morphology aligned with the minor axis, consistent with dust self-scattering. This region provides a constraint for the dust grain size. The goal of the modelling is to model the dust

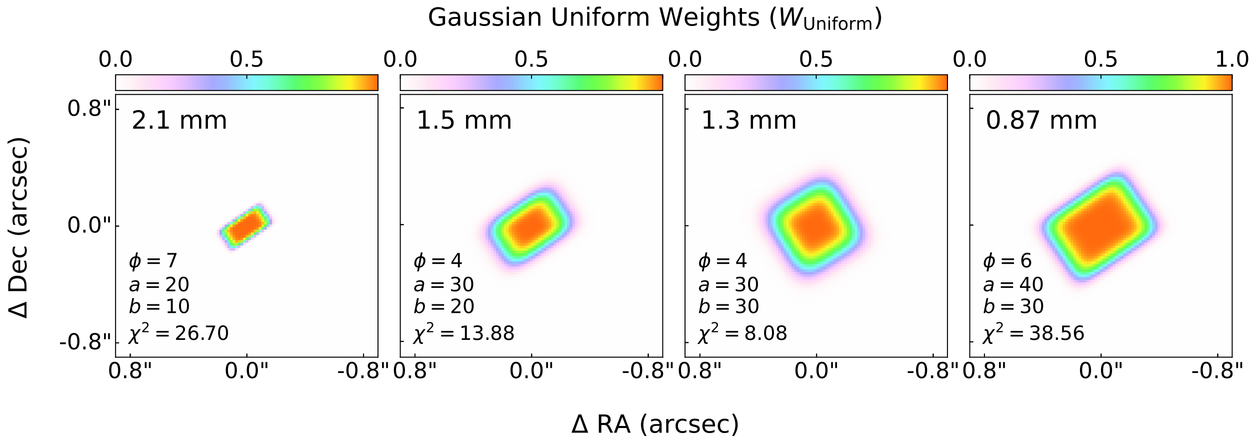

Figure 5.5: Maps of the Gaussian uniform weights W_{uniform} (Equation **??**) for the best-fit Gaussian model for each wavelength, corresponding vector pattern shown in Figure **??**.

properties in this region using spherical and irregular grains and compare the amount of polarized light to the observed polarization fraction of IRS 63 to constrain the dust grain size. Several parameters are used throughout the dust model calculations. Table **??** summarizes these parameters for reference.

5.2.1 Model Assumptions

Dust Composition

The dust grains are assumed to follow the Disk Substructures at High Angular Resolution Project (DSHARP) dust composition, consisting of water ice, astronomical silicates, troilite and refractory organic material (**?**). The corresponding volume fractions, mass fractions, bulk densities, and sources for the refractive indices are listed in Table **??**.

Table 5.4: Summary of parameters used throughout the dust scattering model calculations. question Take any out or add some?

Symbol	**Description**	**Units**
a_{max}	Maximum dust grain size in distribution grid	μm
a	Radius of spherical grain	mm
x	Size parameter of spherical grain	–
λ	Observing wavelength	mm
$\mathcal{P}$	Dust grain porosity	–
f	Dust grain porosity (filling factor)	–
n	Real part of the refractive index	–
k	Imaginary part of the refractive index	–
ω, ω_{eff}	(effective) Single scattering albedo	–
Θ	Scattering Angle	$^\circ$
P	Polarization degree	–
$P\omega_{\mathrm{model}}$ *[NOTE: change??]*	Model polarization fraction	–
$\mathcal{P}_{F,\mathrm{obs}}$	Observed polarization fraction	–
SF	Scale factor	–
χ^2_{P}	Chi-squared statistic for the polarization fraction fit	–

Note: λ and a are reported in mm, while a_{max} is reported in μm. All three are converted to cm for the calculations.

Table 5.5: DSHARP dust composition used in this project **?**.

Material	**Bulk Density (g cm^{-3})**	**Volume Fraction**	**Mass Fraction**	**Refractive Index Source**
Water Ice	0.92	0.3642	0.2000	**?**
Astronomical Silicates	3.30	0.1670	0.3290	**?**
Troilite	4.83	0.02578	0.07434	**?**
Organics	1.50	0.4430	0.3966	**?**

Dust Size Distribution

The dust grains were assumed to have a size distribution of $n(a) \propto a^{3.5}$, chosen to match the distributions used by **?** and **?**. The minimum grain size is 0.1 μm, while the maximum grain size is varied in logarithmic space up to 5000 μm, with 2000 values sampled in total. The maximum grain size in each distribution is denoted by a_{max}. For the irregular grain models, `glitterin` produced Nan values at the lower limit of 0.1 μm, so a lower bound of 2 μm was used.

5.2.2 Spherical Dust Grains

Geometry & Mie Theory

Spherical dust grains were studied first because their scattering properties are less computationally expensive to calculate. The size of a spherical dust grain is defined by its radius (a), with a corresponding size parameter (x) given by:

$$x = \frac{2\pi a}{\lambda}, \tag{5.11}$$

where λ is the observing wavelength (**?**). When the size parameter is equal to 1, this is defined as the characteristic grain size (a_{char}), equal to $\lambda/2\pi$.

The size parameter controls the way the light interacts with the dust grains by determining the scattering regime of the dust grains (Mie theory):

1. $x \ll 1$: Rayleigh regime,
2. $x \sim 1$: Mie regime,
3. $x \gg 1$: geometric regime,

(**?**). The dust models were calculated using the DSHARP Mie-Opacity Library (`DSHARP`) from **?**.

Porosity/Filling Factor and Density

Porosity ($\mathcal{P}$) is the fraction of a dust grain's volume that is empty space. A low porosity corresponds to a more compact grain (like a rock), while a high porosity corresponds to a grain with more empty space (like a sponge).

Instead of specifying the porosity directly, the filling factor (f) is used, defined as:

$$f = 1 - \mathcal{P}, \tag{5.12}$$

where $f = 1$ corresponds to a compact grain ($\mathcal{P} = 0$), and an f value less than 1 corresponds to a more porous grain (**?**). A range of filling factors was tested to sample different dust grain porosities, and four values are presented in Chapter **??**: $f = 1, 0.5, 0.3, 0.1$.

For each filling factor, the `DSHARP` code was used to calculate the corresponding density and optical constants of the dust mixture. The resulting density values are shown in Table **??**. As expected, the density decreases as the filling factor decreases and the porosity increases.

Q: Add more detail on how the `DSHARP` code incorperates different filling factors?

Table 5.6: Summary densities calculated using `DSHARP` for each filling factor.

Filling Factor (f)	**Density (g/cm^3)**
1.0	1.675
0.5	0.838
0.3	0.503
0.1	0.168

Complex Refractive Index

The complex refractive index (m) describes the optical properties of the dust and is given by:

$$m(\lambda) = n(\lambda) + ik(\lambda), \tag{5.13}$$

where n and k are the real and imaginary parts of the refractive index, respectively (?).

Computing Optical Properties

For each filling factor and grain size in the distribution, `DSHARP` calculates several optical properties. These include the mass absorption and scattering coefficients, κ_{abs} and κ_{sca}, respectively. These coefficients describe how efficiently a material absorbs and scatters light. The resulting quantities are averaged over the grain-size distribution and are used to calculate additional parameters described below.

Scattering can have a bias towards the forward direction, which is accounted for using g, the forward-scattering parameter defined by the expectation value of $\cos\Theta$ (?). The scattering angle (Θ) is the angle between the original direction and the new direction after it becomes polarized (?). The parameter g is used to calculate an effective scattering opacity, which is calculated by:

$$\kappa_{\text{sca,eff}} = (1 - g)\kappa_{\text{sca}}. \tag{5.14}$$

An additional quantity is the single scattering albedo, which describes the fraction of the original light that is removed by scattering rather than absorption. The standard and effective albedo are defined as:

$$\omega = \frac{\kappa_{\rm sca}}{\kappa_{\rm abs} + \kappa_{\rm sca}}, \quad \omega_{\rm eff} = \frac{\kappa_{\rm sca,eff}}{\kappa_{\rm abs} + \kappa_{\rm sca,eff}}, \tag{5.15}$$

(?).

Mueller Matrix & Computing Polarization, P

The Mueller matrix, or scattering matrix, describes the polarization properties of light scattered by the dust grains (?) and is calculated using `DSHARP`. Two relevant matrix elements are Z_{11} and Z_{12}, which represent linear polarized scattering and total scattering, respectively (?). The degree of polarization is given by the ratio $-Z_{12,\rm eff}/Z_{11,\rm eff}$ averaged over the grain size distribution (?). The polarization degree depends on the scattering angle, as shown in Figure ??. For smaller grain sizes, the polarization degree peaks near a scattering angle of 90°. The polarization degree at 90° is denoted by P.

Pω distribution

The product of the polarization degree and single-scattering albedo, $P\omega$, provides a measurement of the contribution of each grain size to the polarized scattered light at a specific observing wavelength. For single scattering, $P\omega_{eff}$ can be used to determine the grain sizes that contribute the most to the observed polarization (?). The $P\omega_{eff}$ factor as a function of grain size is shown in Figure ??. In spherical modelling, $P\omega$ denotes $P_{eff} \times \omega_{eff}$.

5.2.3 Irregular Dust Grains

Spherical dust grain modelling is a more common approach in the field because the calculations are less computationally expensive. However, as mentioned previously, non-spherical dust grains are thought to be a realistic representation of dust grains in disks. The change in grain shape affects the scattering properties, making the calculations more difficult. Non-spherical (irregular) dust grain modelling has become more accessible with `glitterin` from **?**, a neural network trained to predict the light-scattering properties of irregular dust grains. An example of six irregular grain shapes is shown in Figure **??**. For the irregular dust grain models, the porosity cannot be varied, so four models were tested, compared to 16 total for the spherical models.

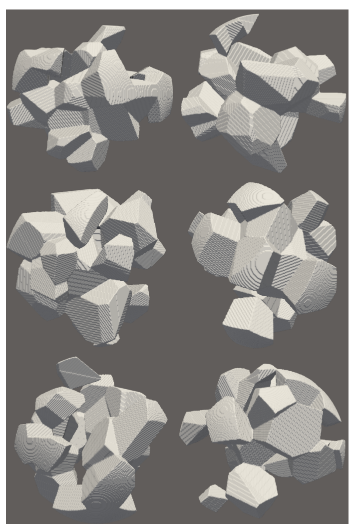

Figure 5.6: Example of six irregular particles from `glitterin`. Figure from **?**. *[NOTE: ask for permission]*

Q: As far as I know, glitterin is using ω and not ω_{eff}, meaning when I present the results for spherical vs irregular, I'm comparing $P\omega_{eff}$ vs $P\omega$. I am not entirely sure how to get the g factor I need to calculate ω_{eff} with gitterin. I only switched to ω_{eff} because of **?**. Should I switch the spherical back to just ω so they are comparing the same thing?

Q: How much detail do I go into about glitterin?

5.3 Replicating Previous Work for Spherical Grains

To verify the spherical dust-grain modelling, several plots from **?** were replicated. These comparisons provide a check that the calculated optical properties are in agreement with previously published results.

5.3.1 Dust Opacities as a Function of Observing Wavelength

Figure **??** shows a replication of Figure 1 from **?**. This figure shows the absorption and scattering mass capacities, in units of cm^2 per gram, as a function of observing wavelength for maximum grain sizes of $a_{\max} = 1$ μm and 100 μm. For the smaller dust grains (red lines, $a_{\max} = 1$ μm), the absorption opacity is much greater than the scattering opacity. For the larger dust grain (blue lines, $a_{\max} = 100$ μm), the scattering opacity is greater than the absorption opacity over approximately 0.1-2 mm. Therefore, at the observing wavelengths in this project, larger grains are more efficient at scattering than smaller grains. At the shorter and longer wavelength bounds, however, the scattering and absorption opacities are relatively similar.

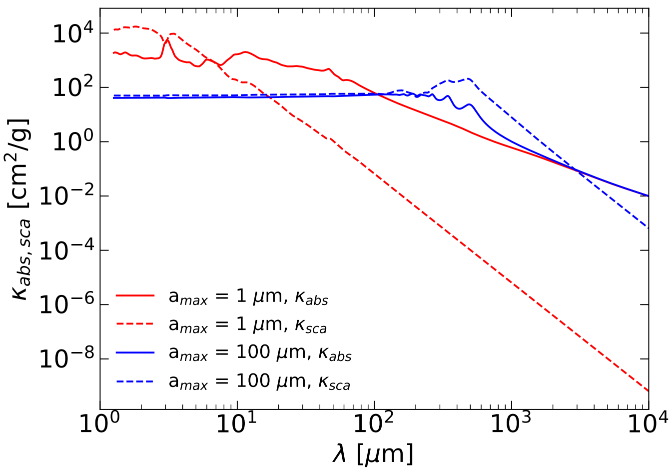

Figure 5.7: Absorption and scattering mass opacities as a function of observing wavelength for grain sizes of 1μm and 100 μm. Recreation of Figure 1 from **?**.

5.3.2 Polarization Dependence on Scattering Angle

Figure **??** shows a replication of Figure 2 from **?** for the four wavelengths considered in this project. The figure shows the dependence of polarization degree on scattering angle for different maximum grain sizes. For smaller dust grains (10 μm and 100 μm), the polarization degree peaks near a scattering angle of 90 °. For larger dust grains (1 mm and 1 cm), the polarization degree remains much closer to zero, indicating that polarization from scattering is not efficient at larger grains.

5.3.3 Polarization and Albedo as a Function of Grain Size

Figure **??** shows a recreation of Figure 3 from **?** for the four wavelengths considered in this project. The figure shows the polarization degree and effective albedo as a function of grain size. The polarization degree is larger for smaller grains, and quickly drops off as the dust grains increase. Alternatively, the effective albedo

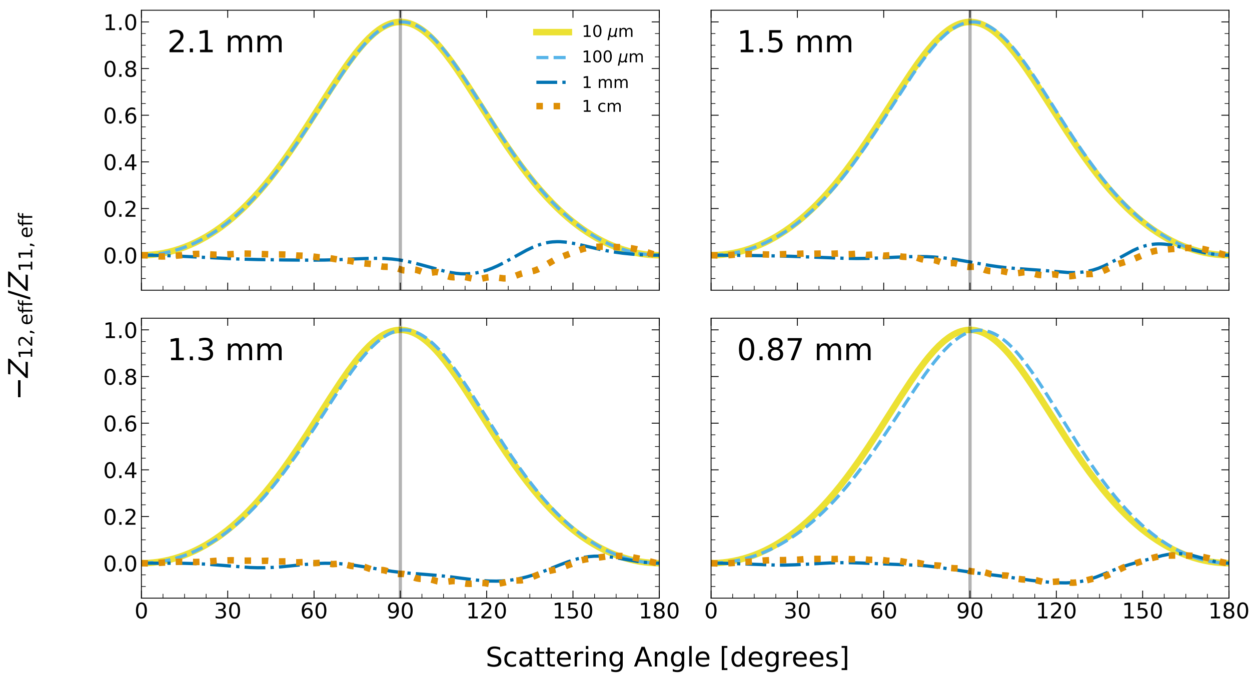

Figure 5.8: Degree of polarization as a function of scattering angle for maximum grain sizes of 10 μm, 100 μm, 1 mm and 1 cm. This is a replication of Figure 2 from **?**.

increases with dust grain size, as there is more surface area for the light to reflect off of. The characteristic grain size for each wavelength is shown as a grey line (values listed in Table **??**). The characteristic grain size should approximately correlate to where the polarization and albedo curves are at their maximum, highlighting the wavelength dependence of polarization scattering.

5.3.4 Pω as a Function of Grain Size

Figure **??** shows a replication of Figure 4 from **?** for the observing wavelengths considered in this project. The product of P and ω_{eff}shown in Figure **??** are combined for this figure. The polarized scattering efficiency peaks at a grain size close to the

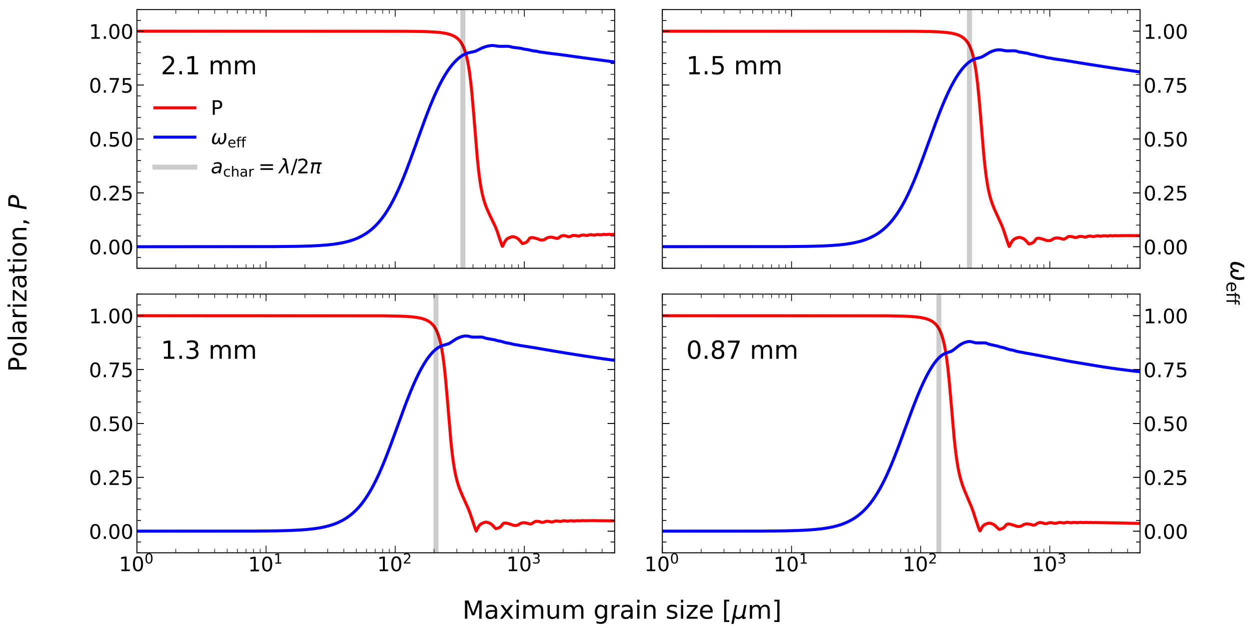

Figure 5.9: Plot of polarization P in red and effective albedo in blue as a function of grain size. The characteristic grain size for each wavelength is overplotted in grey, highlighting that it roughly corresponds to the local maxima of both curves. This is a recreation of Figure 3 from **?**.

characteristic grain size, shown plotted with a dashed line in the respective colour. The peaks occur at grain sizes of approximately hundreds of microns, indicating that polarization observations at these wavelengths are most sensitive to grains in this range. As discussed in Chapter **??**, using multi-wavelength observations allows the different wavelengths to be fit simultaneously, providing a better constraint on the grain size. How this is done is explained in Chapter **??**.

Q: Should I add any replications of **?** plots?

Q: Should I add a section about replicating plots from glitterin? This could be replicating these plots from Kataoka with glitterin, or replicating the plots Daniel presented in his paper.

Table 5.7: Maximum grain size corresponding to the peak $P\omega$ value, and characteristic grain size as a dashed line for each wavelength in Figure **??**. **Q:** Make Figure **??** with glitterin and report that value as well?

Parameter	2.1 mm	1.5 mm	1.3 mm	0.87 mm
$a_{\rm max}$ (μm)	307.88	223.71	196.46	134.50
$a_{\rm char}$ (μm)	334.23	238.73	206.90	138.46

5.4 Measuring Observed Polarization Fraction & Determining the Best-Fit Scale Factor

5.4.1 Measuring Observed Polarization Fraction

A single polarization fraction value was selected at each wavelength for comparison with the dust scattering models. The polarization fraction was selected from the region showing evidence of self-scattering (uniform minor axis alignment). Three methods were tested to determine the most appropriate polarization fraction for this region:

1. $\mathcal{P}_{\rm F,max\ I}$: The polarization fraction at the pixel with maximum Stokes I .

2. $\mathcal{P}_{\rm F,max\,\mathcal{P}_I}$: The polarization fraction at the pixel with maximum polarized intensity.

3. $\mathcal{P}_{\rm F,Gaussian}$: The mean polarization fraction within the uniform area from the Gaussian model ($W_{\rm uniform}$= 1 ± 0.001, Equation **??**).

The values from these three methods at each wavelength are summarized in Table **??**. The best representation of the observations was method 1, where the polarization

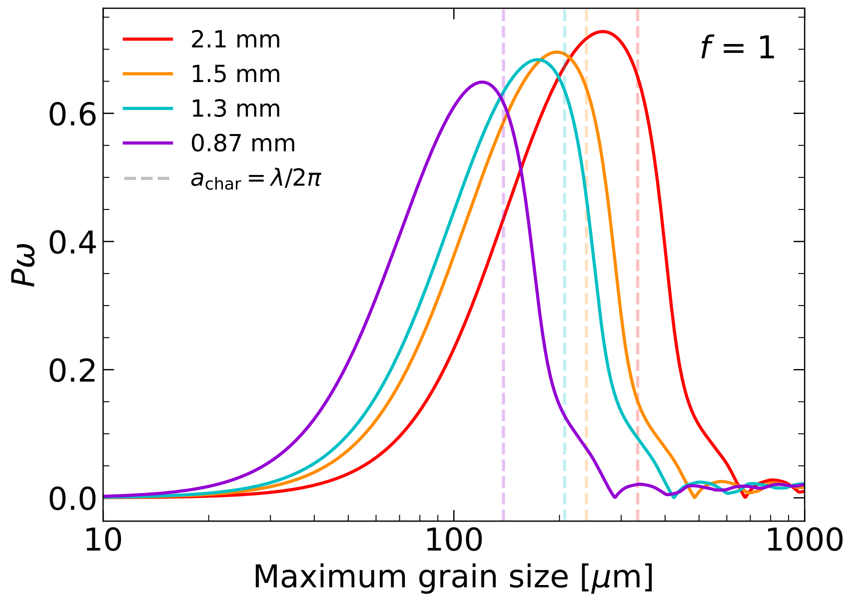

Figure 5.10: Polarization curves ($P\omega$) versus grain size for each wavelength. Overplotted as a dashed line with matching colour is the characteristic grain size. Locations of maximum $P\omega$ and characteristic grain size are listed in Table **??**. This is a recreation of Figure 4 from **?**

fraction was taken at the pixel with the maximum Stokes I value. This selected value is referred to as the observed polarization fraction and is denoted by $\mathcal{P}_{F,\mathrm{obs}}$.

Table 5.8: Results of the three methods tested to find a polarization fraction value in the region where dust self-scattering was observed.

Parameter	2.1 mm	1.5 mm	1.3 mm	0.87 mm
$\mathcal{P}_{\mathrm{F,max\,I}}$	1.37	1.40	1.46	1.04
$\mathcal{P}_{\mathrm{F,max\,POLI}}$	1.56	1.51	1.49	1.08
$\mathcal{P}_{\mathrm{F,Gaussian}}$	1.28	1.19	1.44	0.86

5.4.2 Determining the Best-Fit Scale Factor

The observed polarization fraction ($\mathcal{P}_{F,\mathrm{obs}}$) and the $P\omega$ model values from spherical and irregular grains cannot be directly compared because they represent different quantities. The observed polarization fraction is a dimensionless ratio of polarized intensity to total intensity, while $P\omega$ is the polarization scattering efficiency of the dust. A scale factor (SF) is therefore introduced to relate the model values to observations.

The scale factor is calculated at each a_{max} in the sample, given by:

$$\mathrm{SF}_i(a_{\mathrm{max}}) = \frac{\mathcal{P}_{F,\mathrm{obs,\ i}}(a_{\mathrm{max}})}{P\omega_{\mathrm{model,\ i}}(a_{\mathrm{max}})}, \tag{5.16}$$

where i denotes each wavelength included in the fit. This provides a scale factor for each fitted wavelength at every value of a_{max}. If a given grain size provides a good reproduction of the observations across all fitted wavelengths, the resulting scale factors should be similar between the wavelengths. For each value of a_{max}, the standard deviation of the scale factors is calculated to determine the spread of the scale factors between wavelengths:

$$\mathrm{SF}_{\mathrm{STDEV}} = \texttt{numpy.std}(\mathrm{SF}_i(a_{\mathrm{max}})). \tag{5.17}$$

The best-fit grain size ($a_{\mathrm{max,\ best}}$) is defined as the value of a_{max} where the standard deviation of the scale factors is minimized. The corresponding scale factor is the best-fit scale factor ($\mathrm{SF}_{\mathrm{best}}$). The model $P\omega$ values are multiplied by the $\mathrm{SF}_{\mathrm{best}}$, allowing the model and observations to be directly compared.

Multiple wavelength combinations were tested to determine the effect of each wavelength of data on the success of the fit. The fits were (1) all wavelengths, (2) no 1.5 mm, (3) no 1.3 mm, and (4) just 2.1 mm and 0.87 mm. A chi-squared value was calculated at each fitting combination and porosity to determine the agreement between the model and observations, given by:

$$\chi^2_{\mathcal{P}} = \sum_{\textit{all wavelengths}} \left(\frac{\mathcal{P}_{F,\mathrm{obs}} - [P\omega(a_{\mathrm{max,\ best}}) \times \mathrm{SF}_{\mathrm{best}}]}{\sigma_{\mathcal{P}F}} \right)^2 . \tag{5.18}$$

The calculation is summed over *all wavelengths* as each wavelength was included in the $\chi^2_{\mathcal{P}}$ calculation, regardless of whether it was included in the fit for the scale factor value or not. The fitting combination and porosity that produced the lowest $\chi_{\mathcal{P}}$ value were chosen to be the best-fit model.

5.5 Results from Dust Scattering Models

5.5.1 Spherical Dust Models

The results of the spherical dust models are reported in Table **??**. For each fitting case, the best-fit scale factor from Equation **??**, the corresponding $\chi^2_{\mathcal{P}}$ value (Equation **??**), and the best-fit grain size (a_{max} corresponding to the best SF) are reported. The best fit for each fitting case is shown in bold.

The results of the fits are shown in Figure **??** for the best overall fitting case, which excludes 1.5 mm. Equivalent plots for the other fitting cases are provided in Appendix **??**. The observed polarization fraction, $\mathcal{P}_{F,\mathrm{obs}}$, is shown as a colour-matched marker for each wavelength (values are listed in Table **??**, in the $\mathrm{POLF}_{\mathrm{max\ Stokes}\ I}$

column).

These results show that for all four cases, the lowest $\chi^2_{\mathcal{P}}$ occurs for a filling factor of $f = 0.5$. The compact-grain case of ($f = 1.0$) produces a higher $\chi^2_{\mathcal{P}}$ value in all fitting cases, while the more porous models ($f = 0.3$ and $f = 0.1$) provide worse fits than $f = 0.5$. This suggests that a slightly porous grain model is the best agreement with the observed polarization fraction.

Table 5.9: Results of the four fitting cases for the spherical dust grain modelling. Scale factor calculated using Equation **??**, $\chi^2_{\mathcal{P}}$ from Equation **??** and best dust grain size. The best fit for each fit case is highlighted in bold text.

Fit Case	**Porosity (f)**	**Scale Factor (SF)**	$\chi^2_{\mathcal{P}}$	$a_{\max}$ (μm)
All wavelengths	1.00	2.24	28.65	162
	0.50	**2.13**	**1.31**	**244**
	0.30	2.26	2.64	287
	0.10	2.95	6.13	362
No 1.5 mm	1.00	2.38	34.03	163
	0.50	**2.15**	**1.26**	**245**
	0.30	2.27	2.13	290
	0.10	2.97	5.60	362
No 1.3 mm	1.00	2.38	34.03	163
	0.50	**2.13**	**1.35**	**244**
	0.30	2.26	2.57	289
	0.10	2.94	6.70	364
Just 2.1 mm and 0.87 mm	1.00	2.49	69.98	164
	0.50	**2.15**	**1.44**	**245**
	0.30	2.26	2.32	289
	0.10	2.95	6.54	366

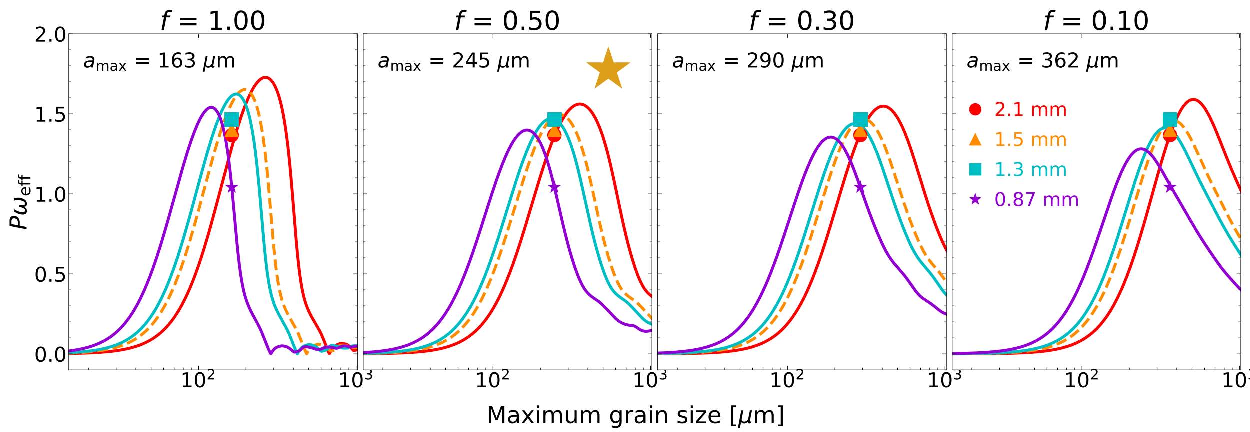

Figure 5.11: Curves of $P\omega$ multiplied by best-fit SF (Equation **??**) to compare to $\mathcal{P}_{F,\rm obs}$. Markers are $\mathcal{P}_{F,\rm obs}$, listed in Table **??**. The best-fit (lowest χ^2, Equation **??**) is represented with a gold star. The specific results of the fit can be found in Table **??**. Wavelengths that are not included in the fit for the SF have a dashed line. **Q:** Q1: Make markers just the outline if not in the fit? Or just a dashed line, okay? Q2: Should I make the x-axes different? As in, at lower f values, should I start at a higher xmin and go to a higher xmax so you can see the entire curve better?

5.5.2 Irregular Dust Models

The results of the four fits for the irregular dust grains are presented in Table **??** and Figure **??**. The lowest $\chi^2_{\mathcal{P}}$ occurs for the no 1.3 mm fit case, however, the all-wavelengths case gives a very similar result. Both cases have a best-fit grain size of 212 μm.

Table 5.10: Same as Table **??** but for irregular grains. Note that for the irregular grains, porosity is not varied.

Fit Case	**Scale Factor (SF)**	$\chi^2_{\mathcal{P}}$	$a_{\rm max}$ **(μm)**
All wavelengths	1.77	4.58	212
No 1.5 mm	1.81	6.16	214
No 1.3 mm	**1.79**	**4.39**	**212**
Just 2.1 mm and 0.87 mm	1.83	8.41	215

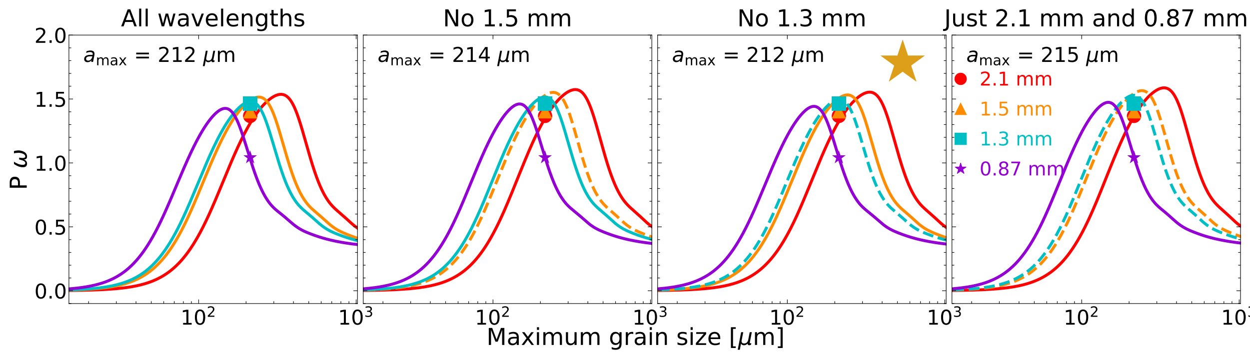

Figure 5.12: Same as Figure **??** but for irregular dust grains. Porosity is not varied, but each fit is represented in the grid. Corresponding results can be found in Table **??**. **Q:** Do I need the titles, or are the dashed lines clear enough? *[NOTE: Legend overlaps with curves a bit; can make text smaller.]*

5.5.3 Comparison of Dust Models

Which geometry gives the better fit? Compare the minimum $\chi^2_{\mathcal{P}}$ values and which fitting case produces them. What grain sizes are favoured? What does the porosity/shape result tell you? For spherical grains, f=0.5 is preferred. For irregular grains, if you have a corresponding best f, compare that. If the irregular models don't have the same interpretation of f, don't force a comparison.

Chapter 6

Discussion

6.1 Dominant Polarization Mechanisms in IRS 63

Mechanism changes with wavelength:

- *"By showing two distinct polarization morphologies and a transition between the two, this paper provides definitive evidence that the dominant (sub)millimeter polarization mechanism transitions with wavelength (**?**)"*.

6.2 Constraints on Dust Grain properties

6.3 Comparison with Previous Studies

6.4 Azimuthal Polarization

6.5 Band 5 0.5 vs -1, streamer, a

6.6 Streamer

6.7 Galaxy Contamination

Chapter 7

Conclusion

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Appendix A

Dust Modeling Images

Q: Do I need to be showing these extra dust model plots? If so, I will add the 'non-best' ones for glitterin as well

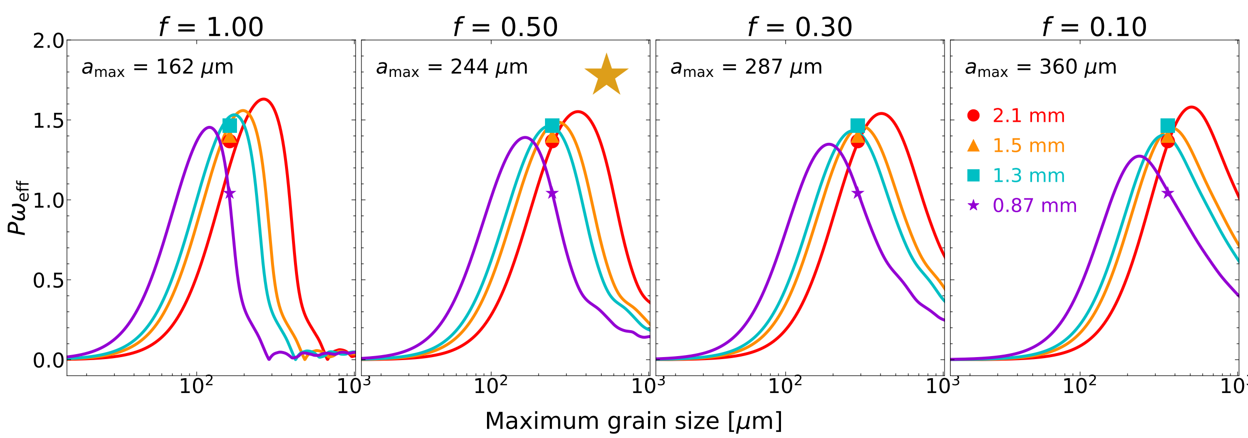

Figure A.1: Same as Figure **??**, but for the all wavelengths fit case.

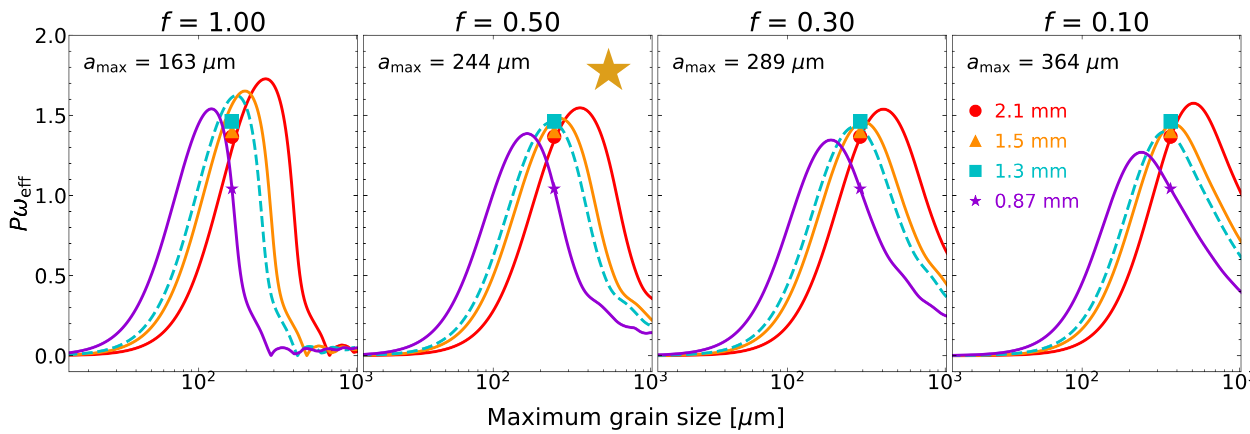

Figure A.2: Same as Figure **??**, but for the no 1.3 mm fit case.

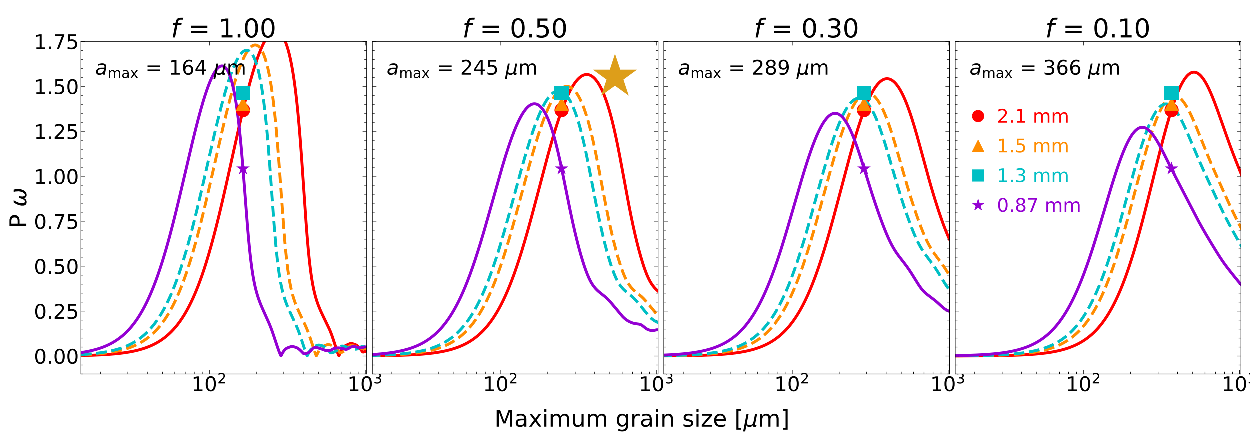

Figure A.3: Same as Figure **??**, but for the just 2.1 mm and 0.87 mm fit case.